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# What's My Client's Wi-Fi Experience?

Amin Sedighfar
CWNE #622

WLPC Prague
OCT 2026



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# Amin Sedighfar

## CWNE #622

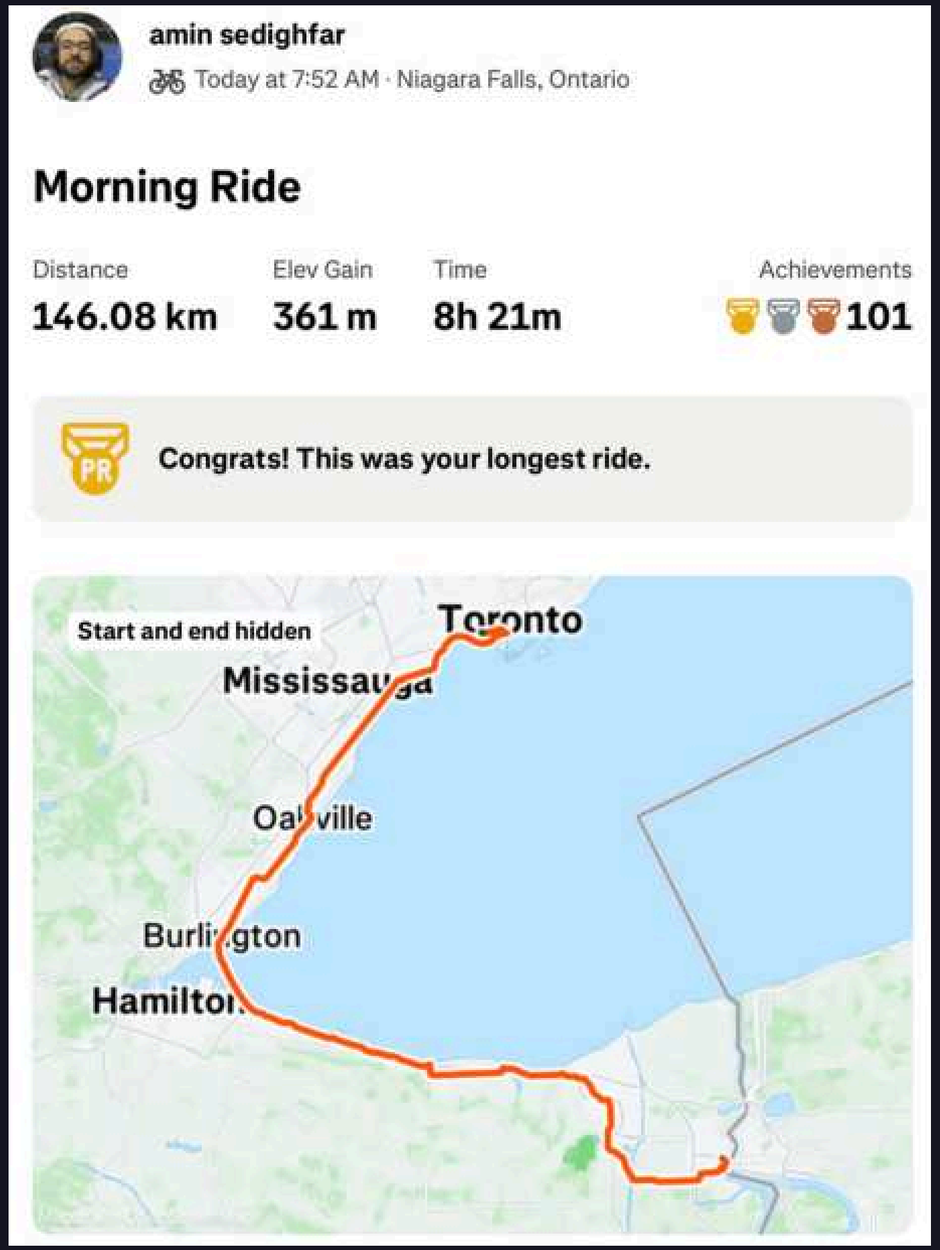



## I Enjoy Learning About Wi-Fi!

## Big fan of 🚴 📶 🥃



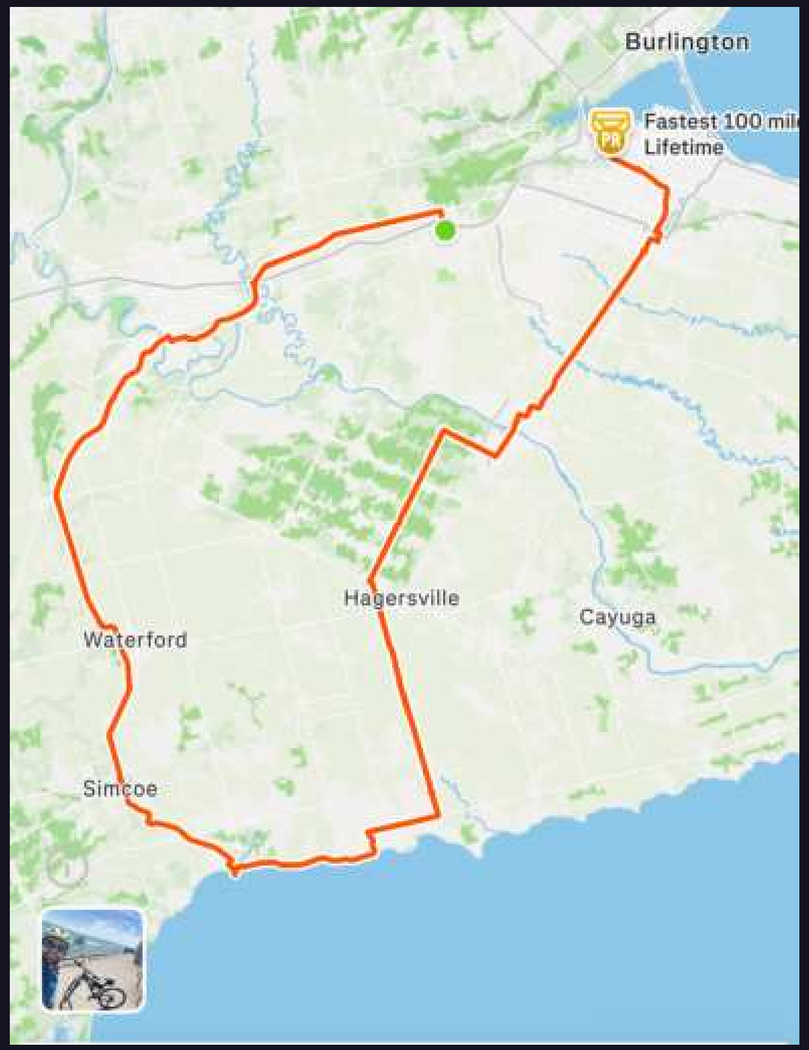


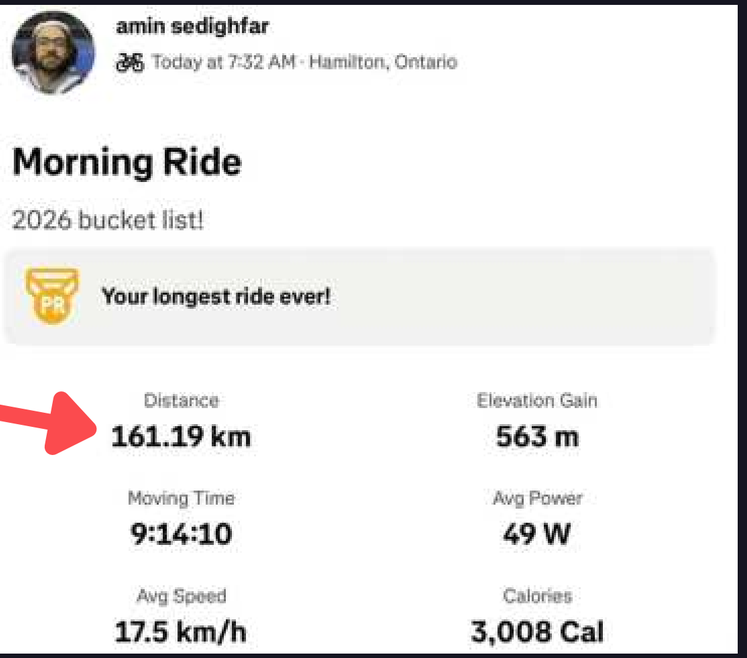

amin sedighfar
Today at 7:52 AM · Niagara Falls, Ontario
Morning Ride
Distance
146.08 km
Elev Gain
361 m
Time
8h 21m
Achievements
101
PR
Congrats! This was your longest ride.
Start and end hidden
Toronto
Mississauga
Oakville
Burlington
Hamilton



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# "Cyclists are better people, but bad for the economy."

Troy Martin

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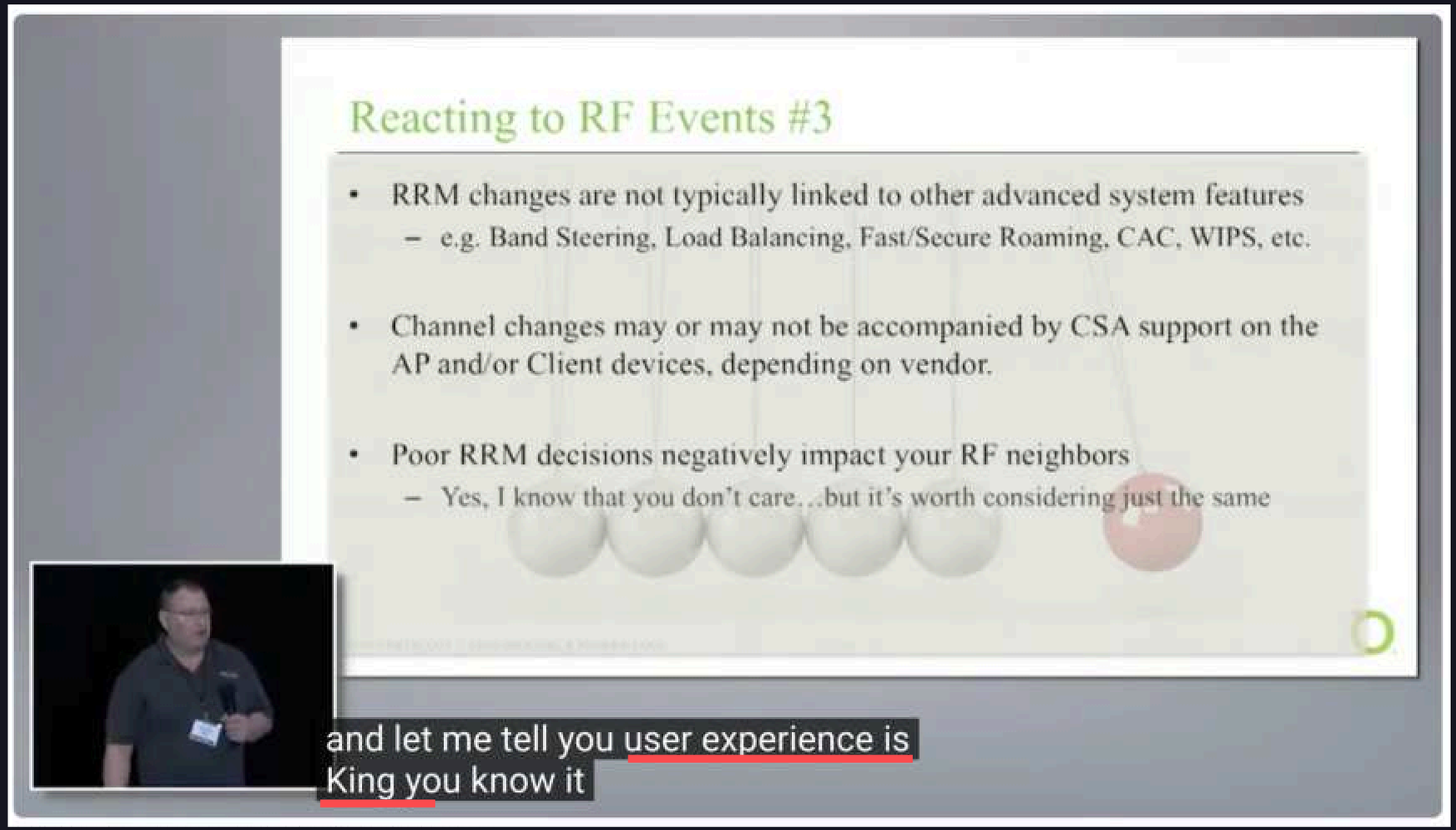
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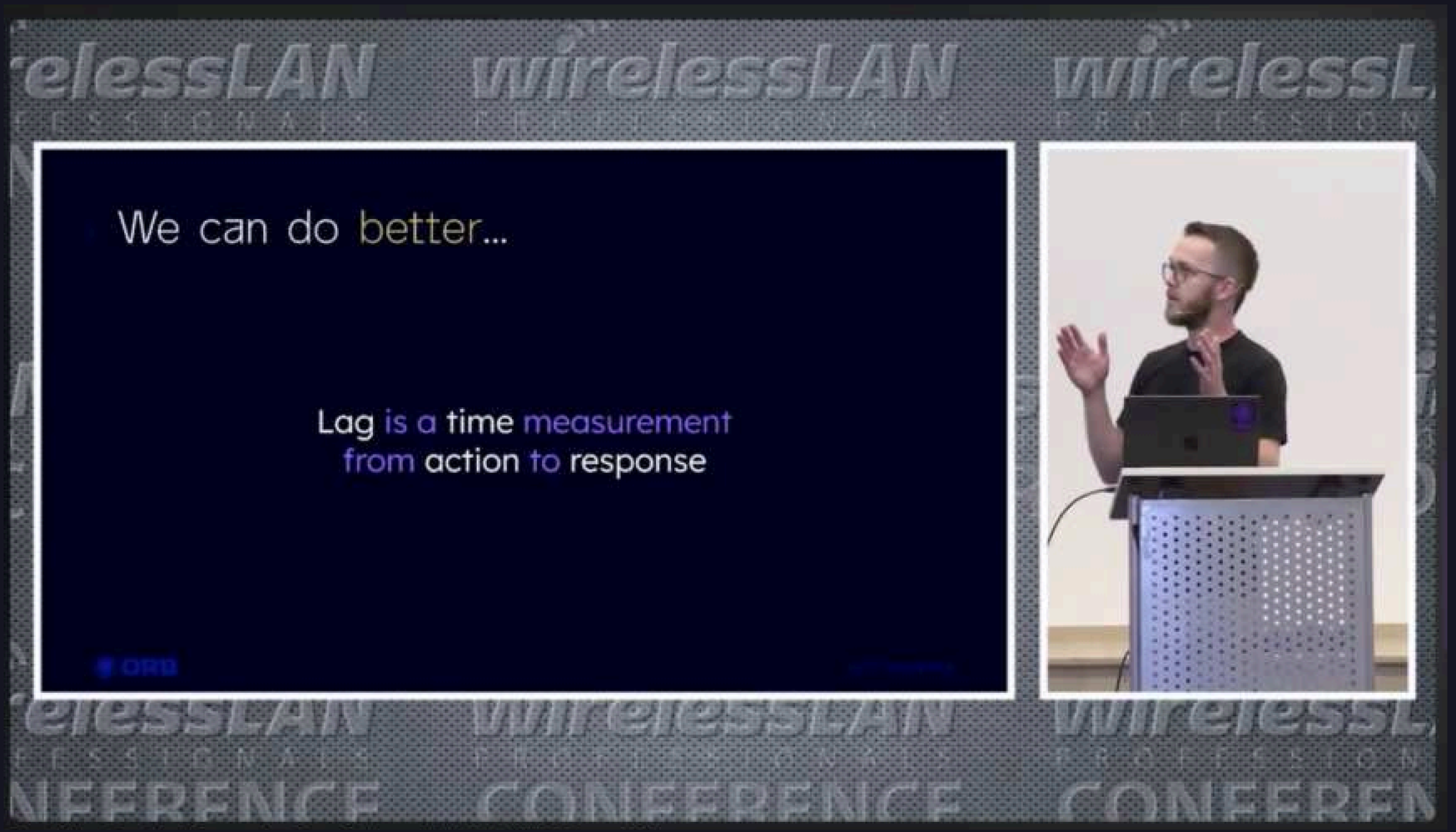
Wi-Fi HaLow 802.11ah & Real-World Performance Results WLPC Phoenix 2024




# "User experience is king"

Devin Akin

Radio Resource Management
WLPC Phoenix 2016



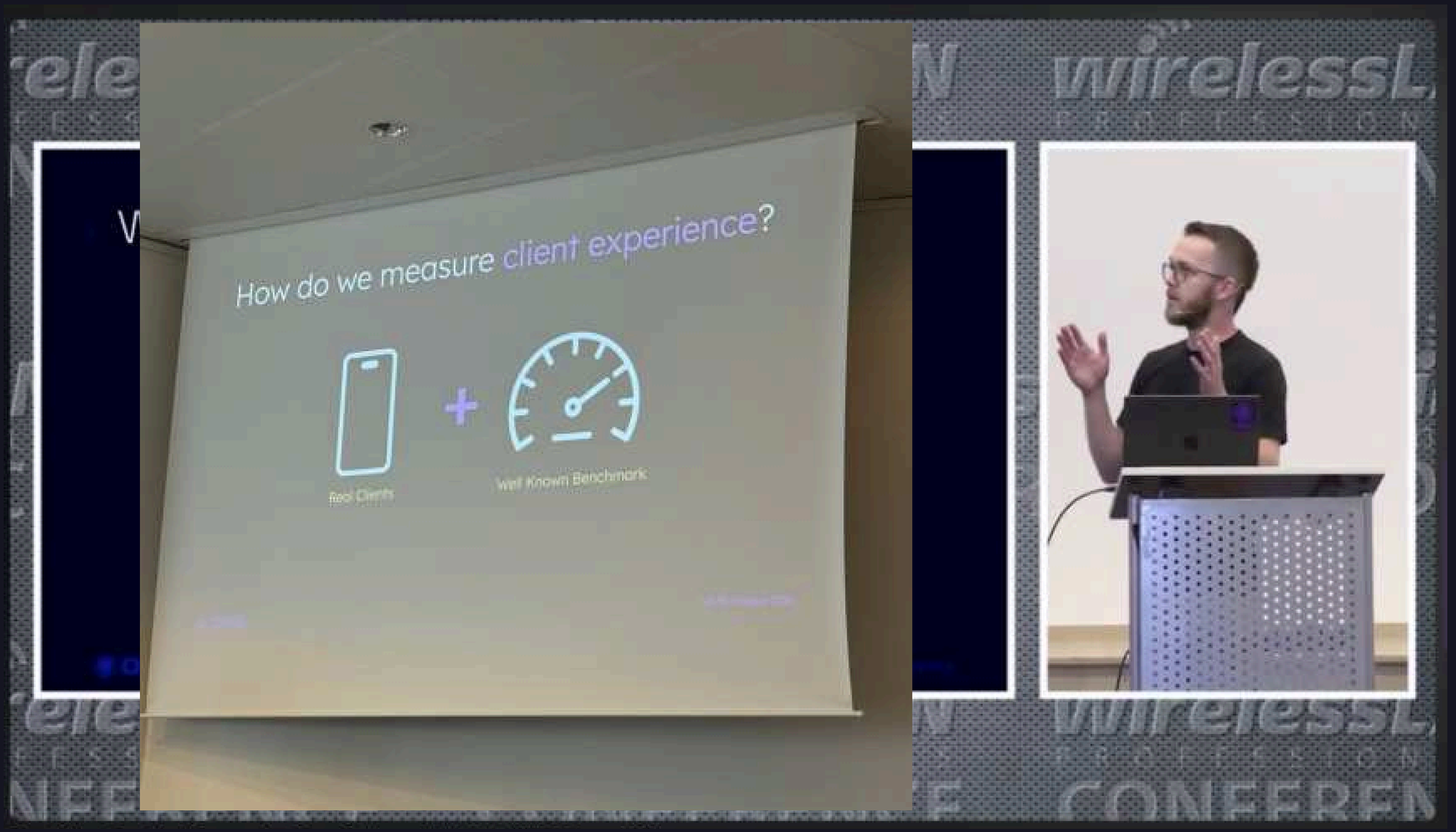

# “We should optimize for lag”

Josh Hardy

Forget About Speed,
Let's Focus on Lag
WLPC Prague 2025

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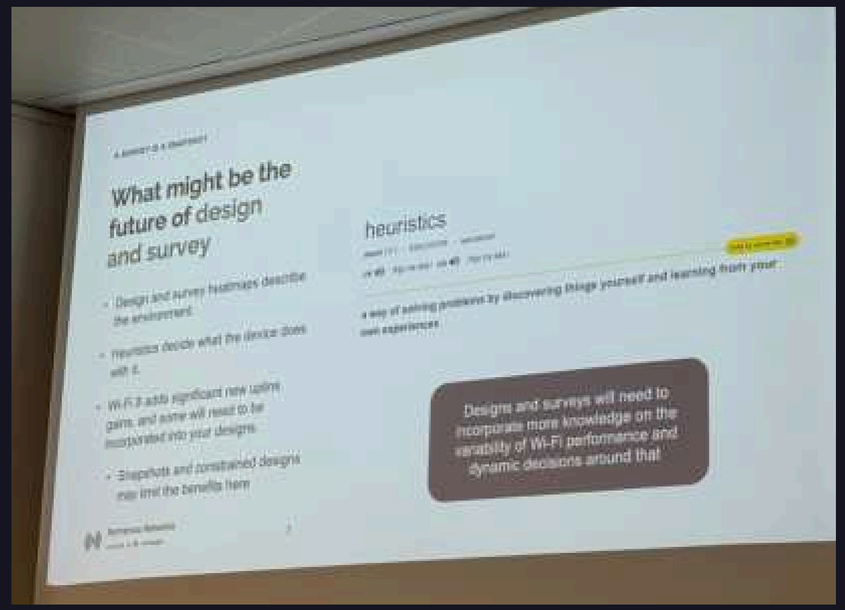
# What's My Client's Wi-Fi Experience?

Not reinventing the wheel!



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# Heuristics:

a way of solving problems by discovering things yourself and learning from your own experiences.



Ben Toner

Founder & CEO of Numerous Networks

# What piece of information do I need?

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# What piece of information do I need?

My client's:

- Received Signal Strength Indicator (RSSI)
- Signal-to-Noise Ratio (SNR)
- Modulation and Coding Scheme (MCS) Index
- And Retry Rate.

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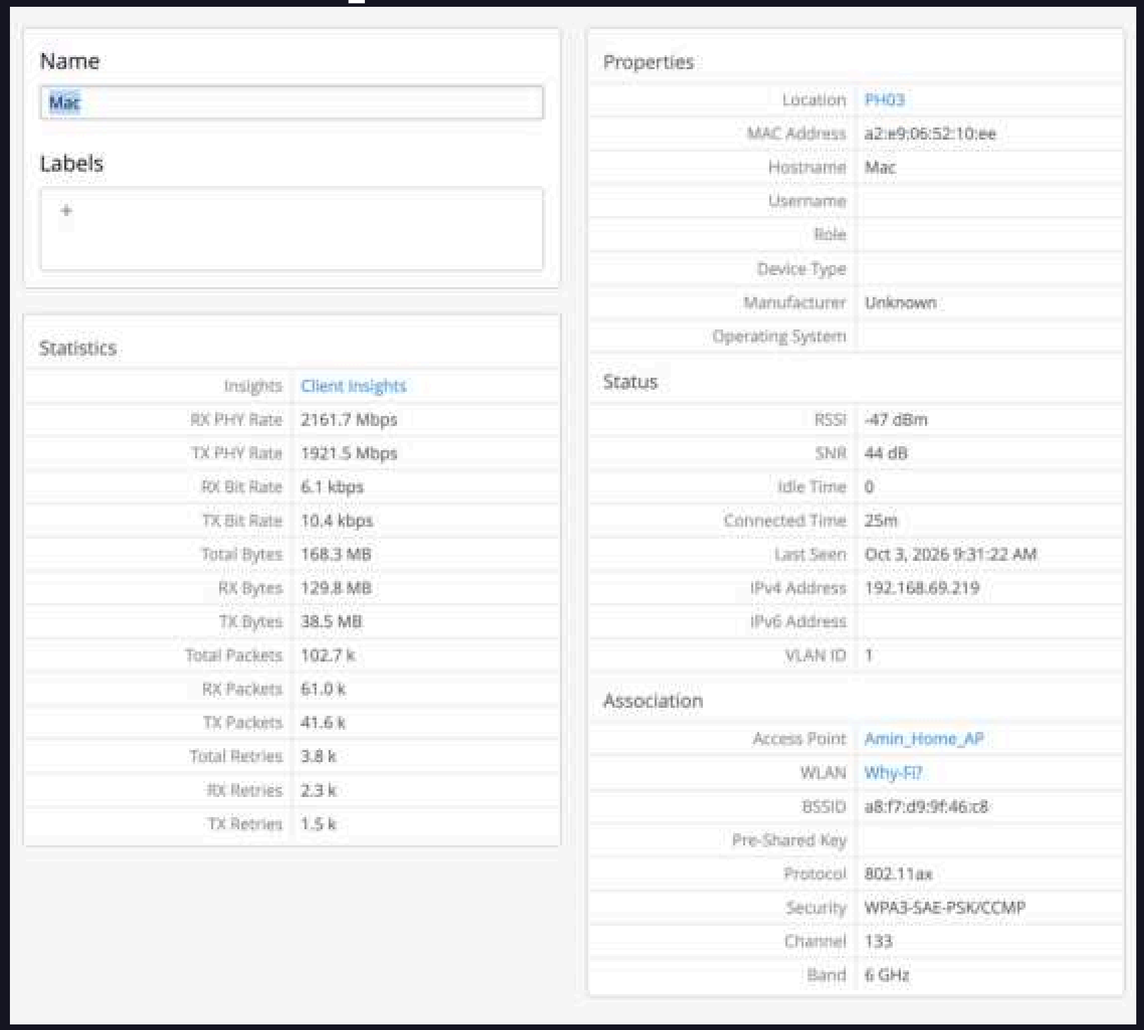


But how do I get these without being physically present? 🤔

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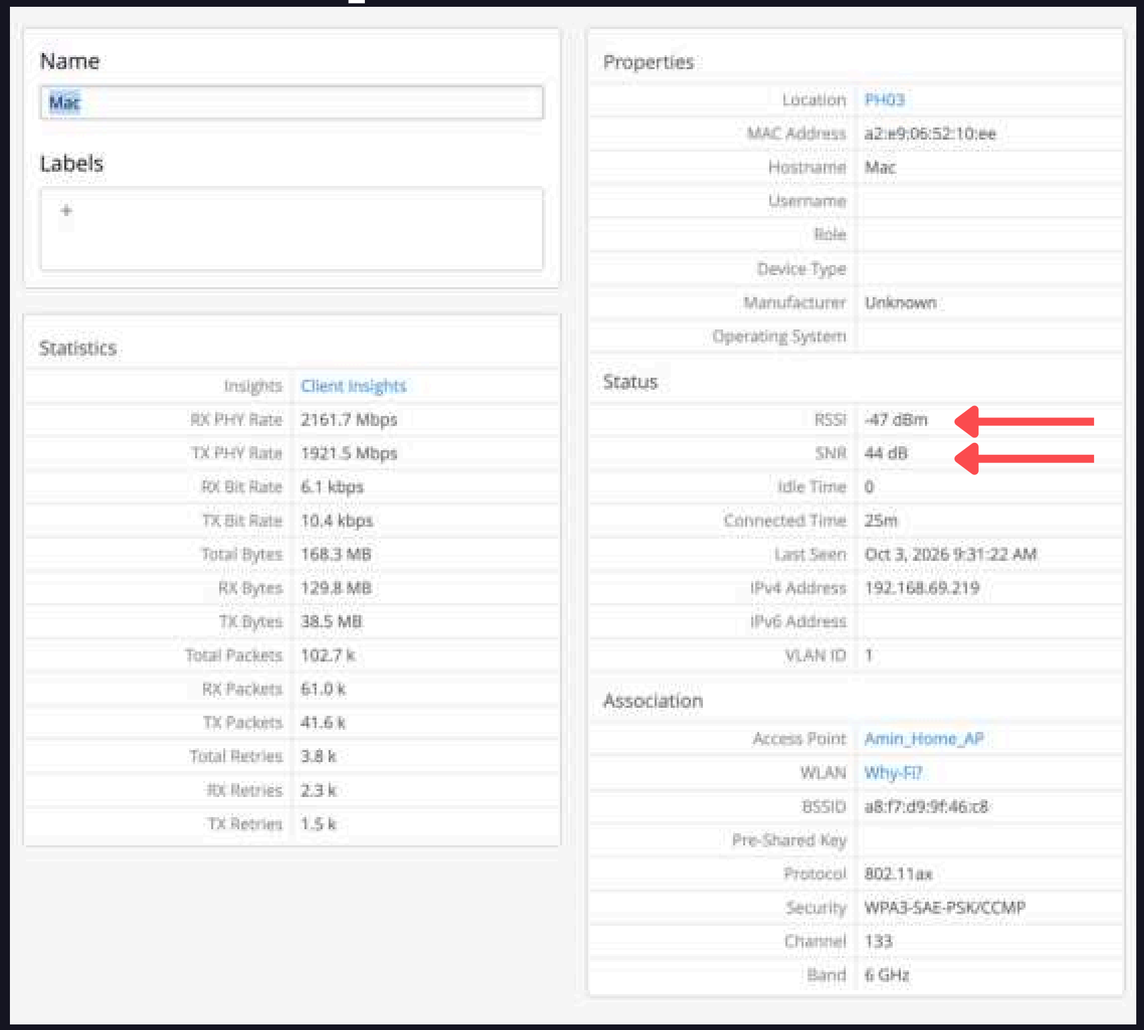



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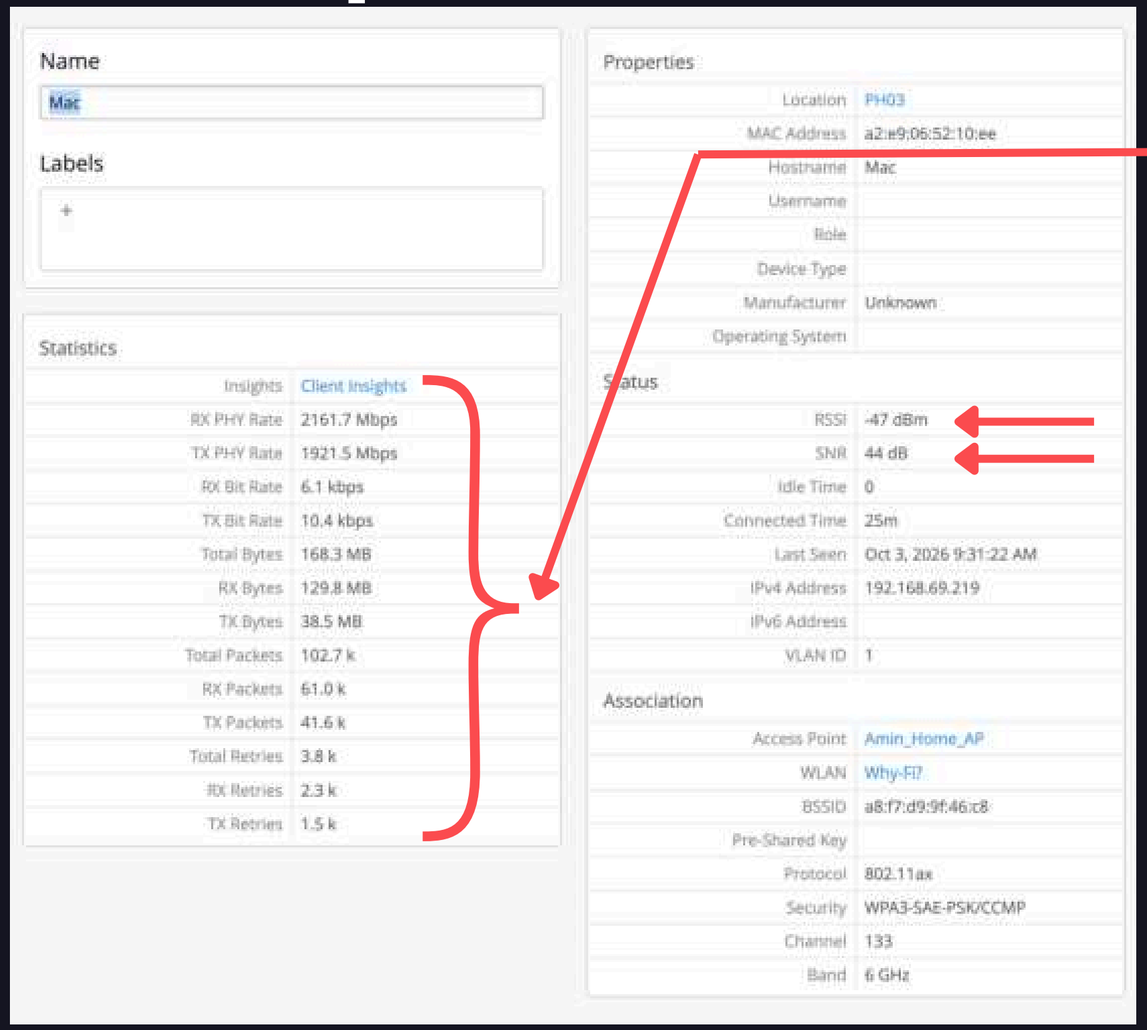
# What piece of information do I need?




**What the AP radio hears on the uplink from the client. It is from the perspective of the receiver.**

**But how do I get these without being physically present?** 🤔

# What piece of information do I need?




Derived from transmitted and received data frames.

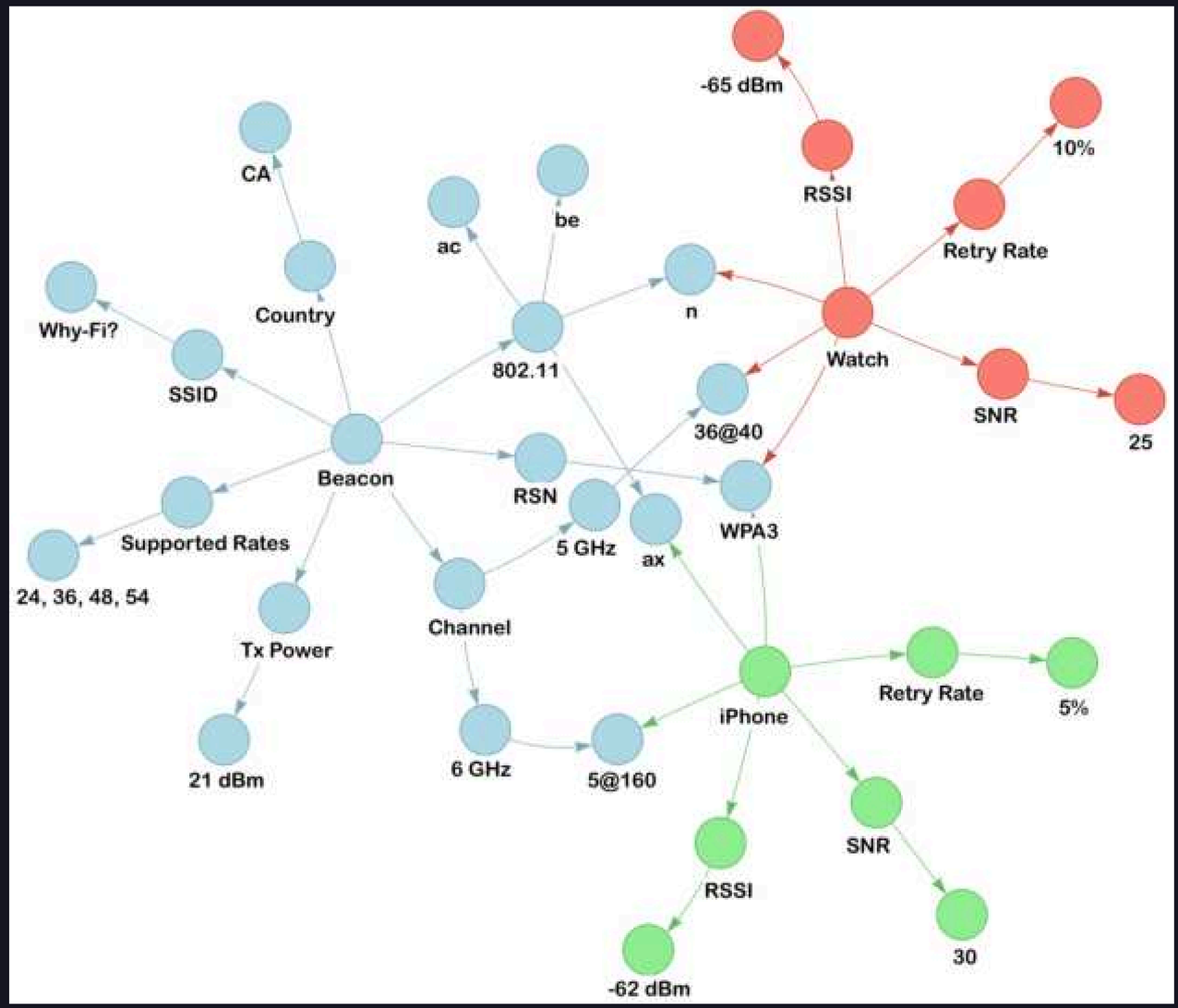
What the AP radio hears on the uplink from the client. So you are correct, it is from the perspective of the receiver.

**But how do I get these without being physically present?** 🤔

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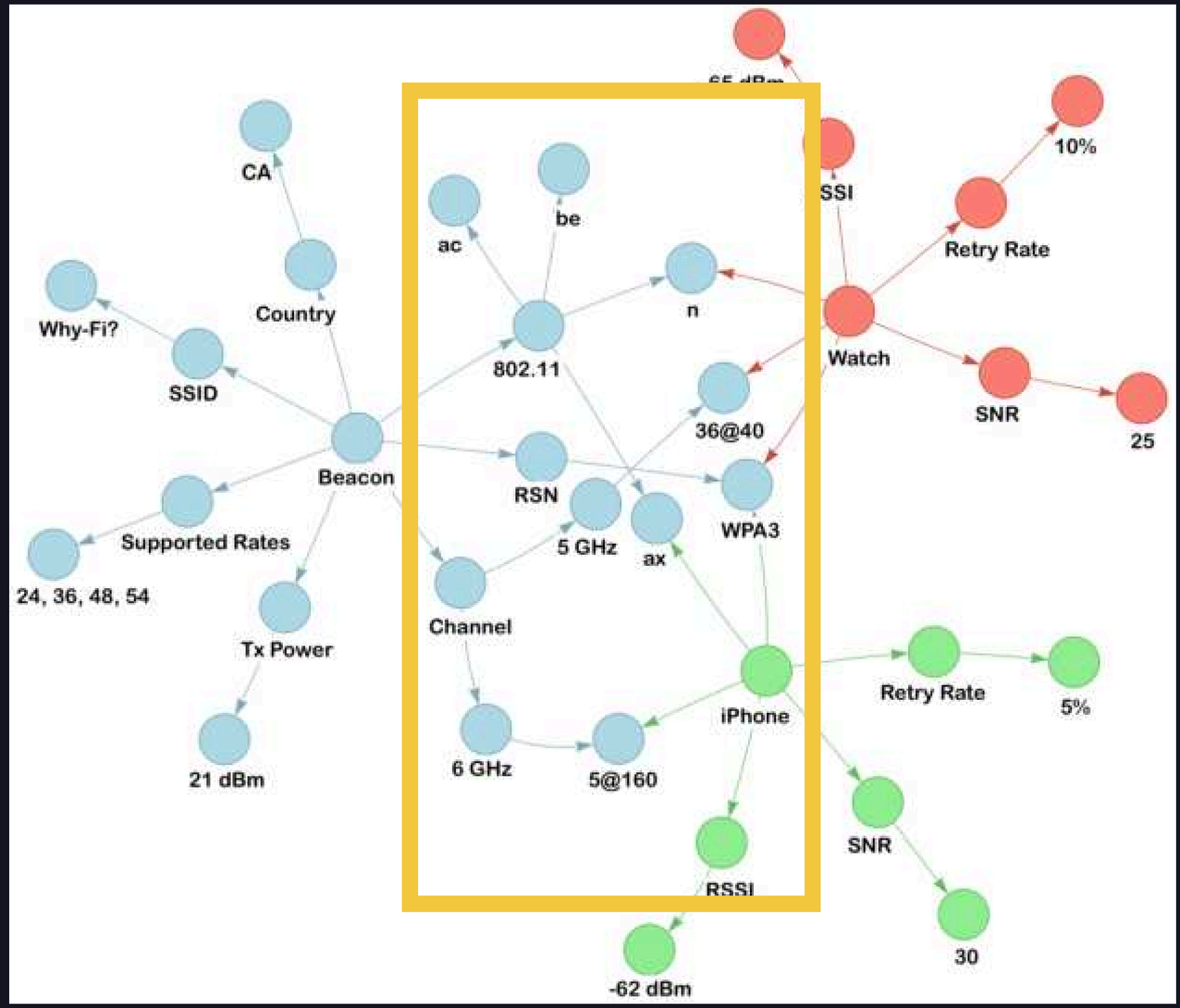
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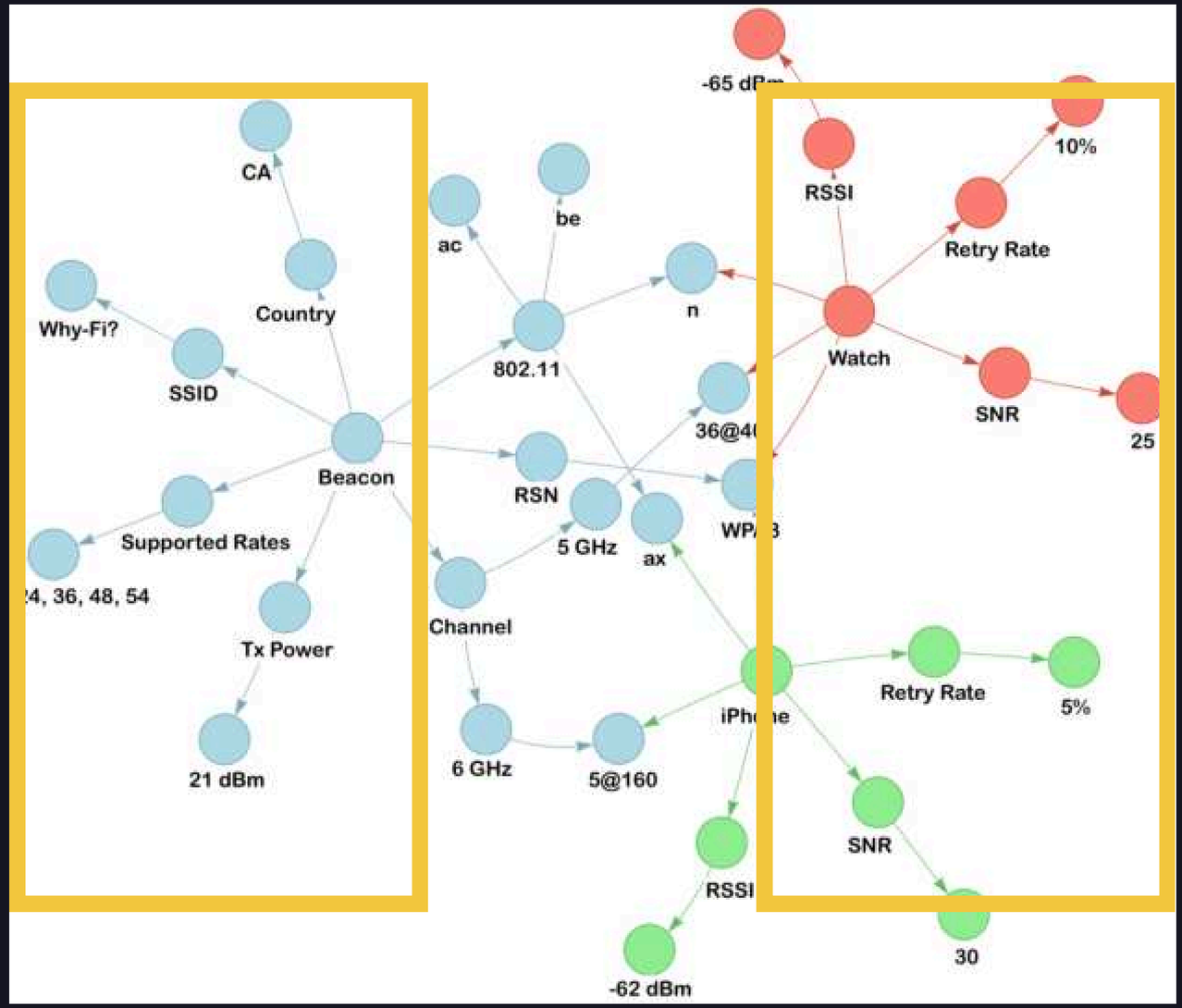
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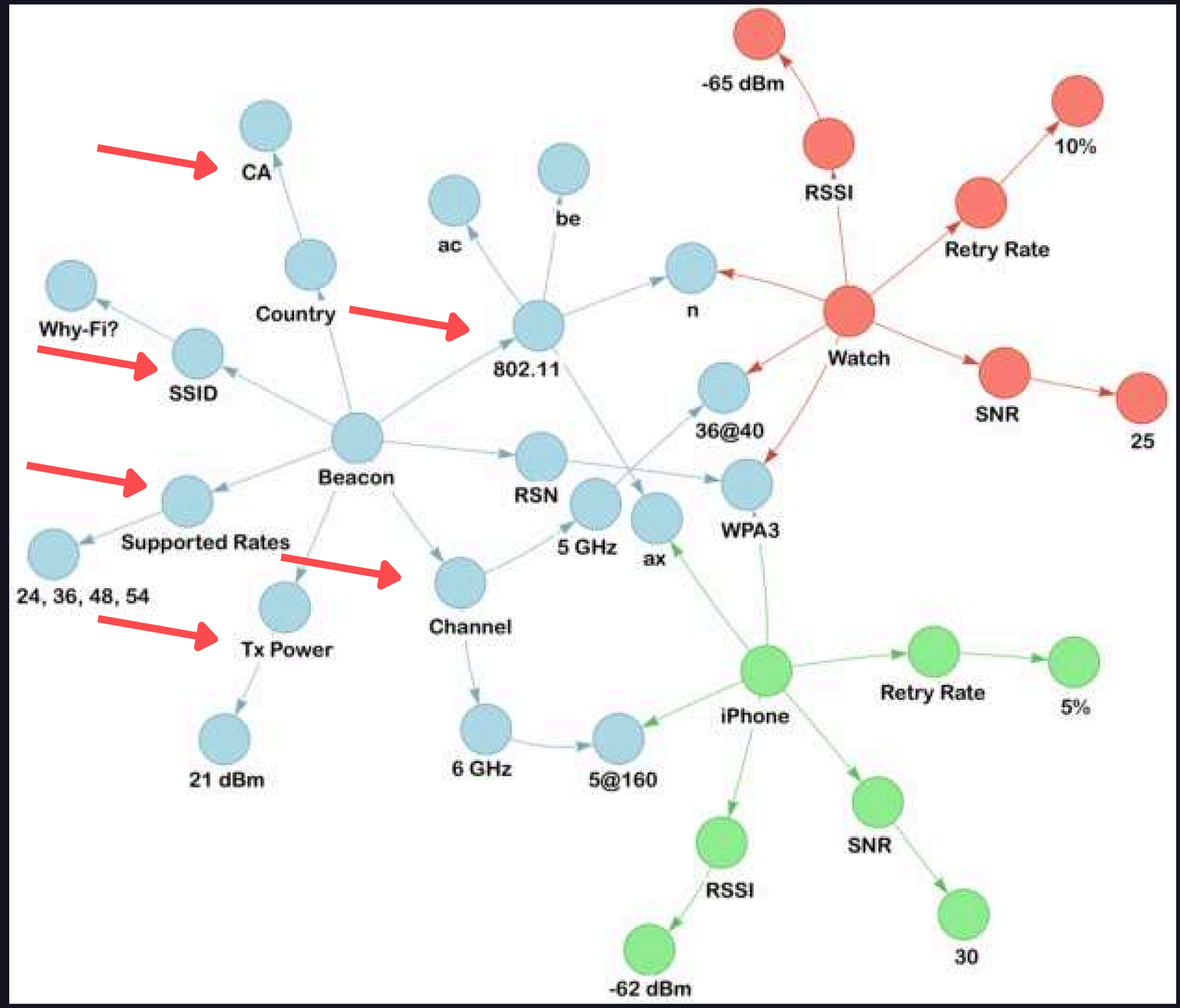


# What piece of information do I need?

IEs coming from the AP:

rf-profile, wlan-template, etc.

This means the AP dictates them.



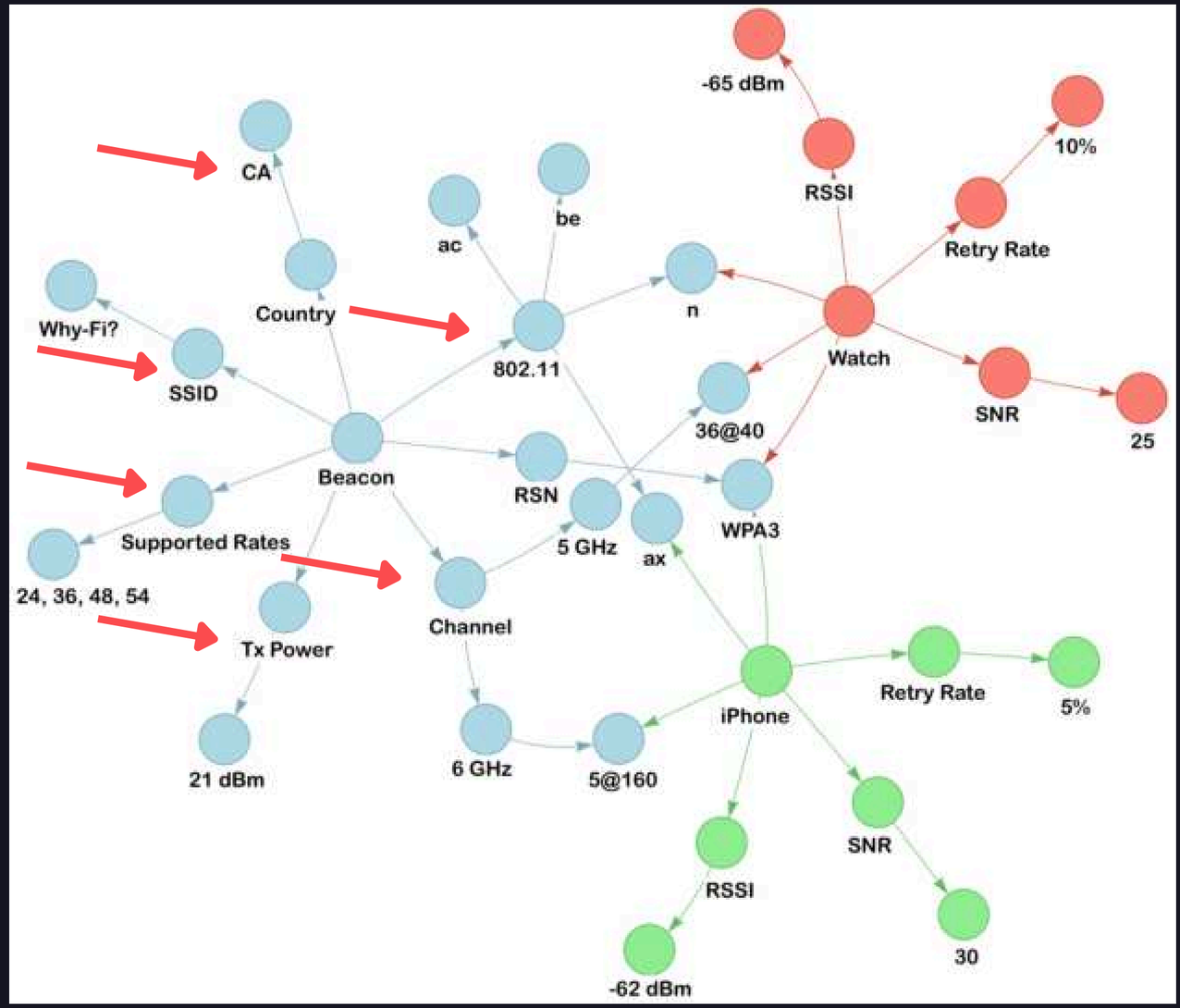
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## Why Are These Important?



Why Are These Important?

Let’s take a quick look at MBR!

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# Why Are These Important?



## Jerry Olla

Head of Technical Stuff

at Hamina Wireless

CWNE #238

At what number do you set your MBR?

At 6, 12, 24, or 54 Mbps?

Data provided by: Jerry Olla

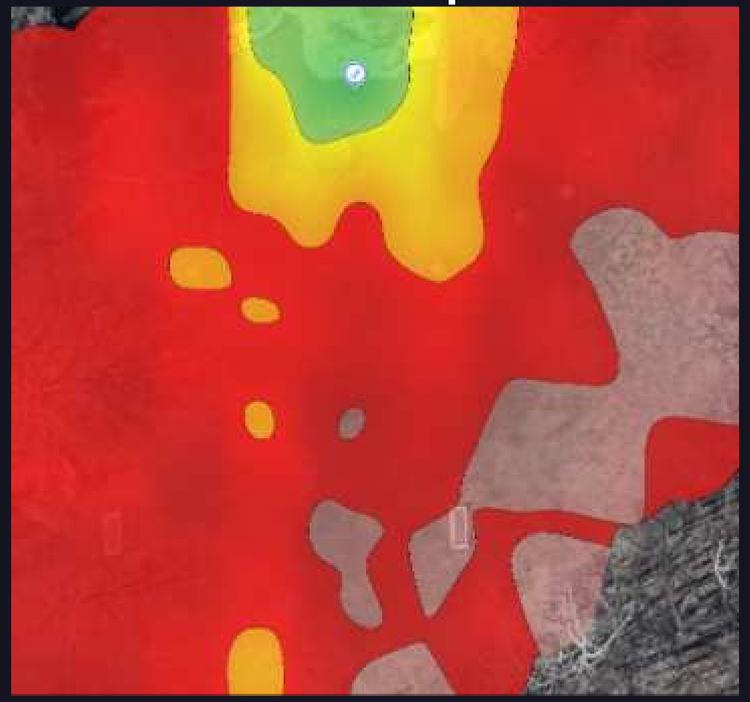
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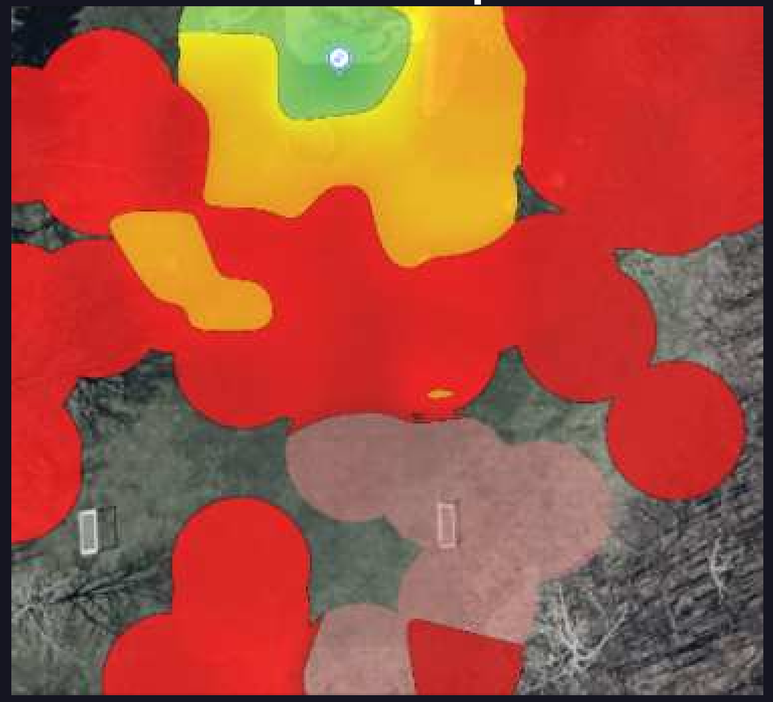
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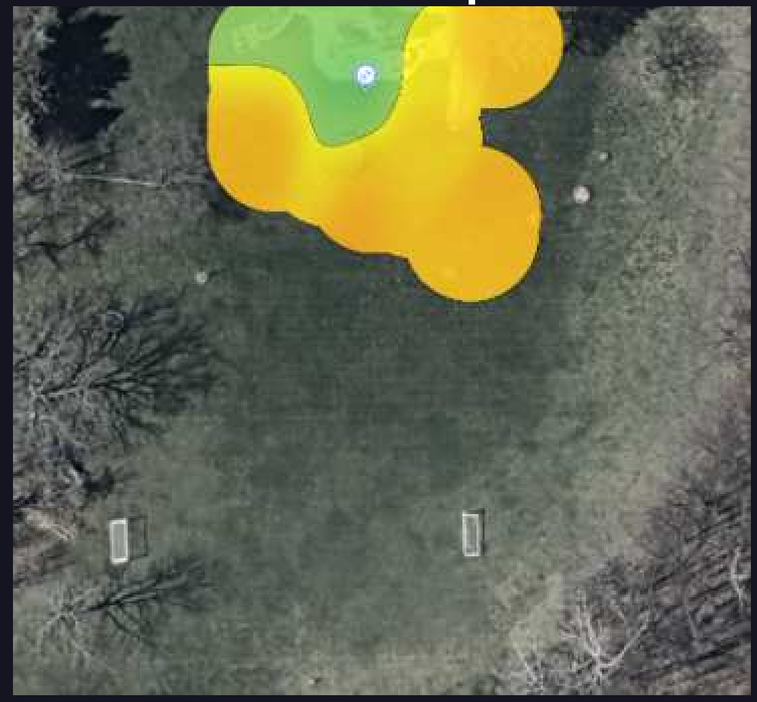
**Min 6 Mbps**



**Min 24 Mbps**



**Min 54 Mbps**



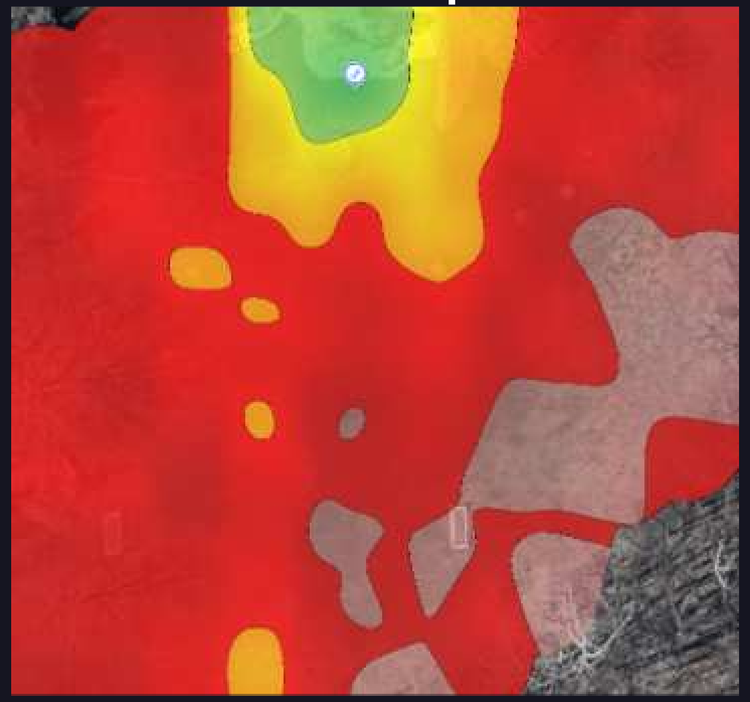
Same AP · Same Power · Same Path · Only minimum data rate changed

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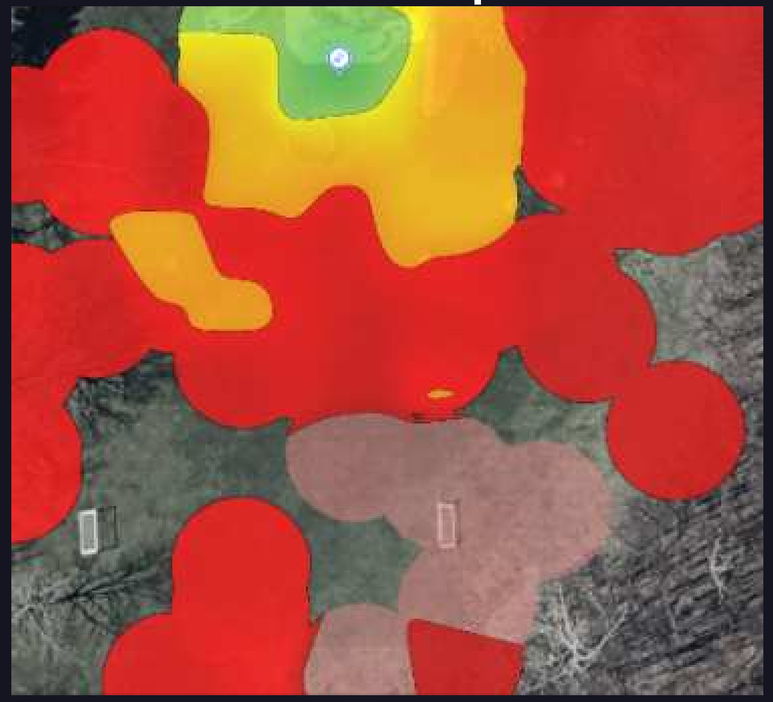
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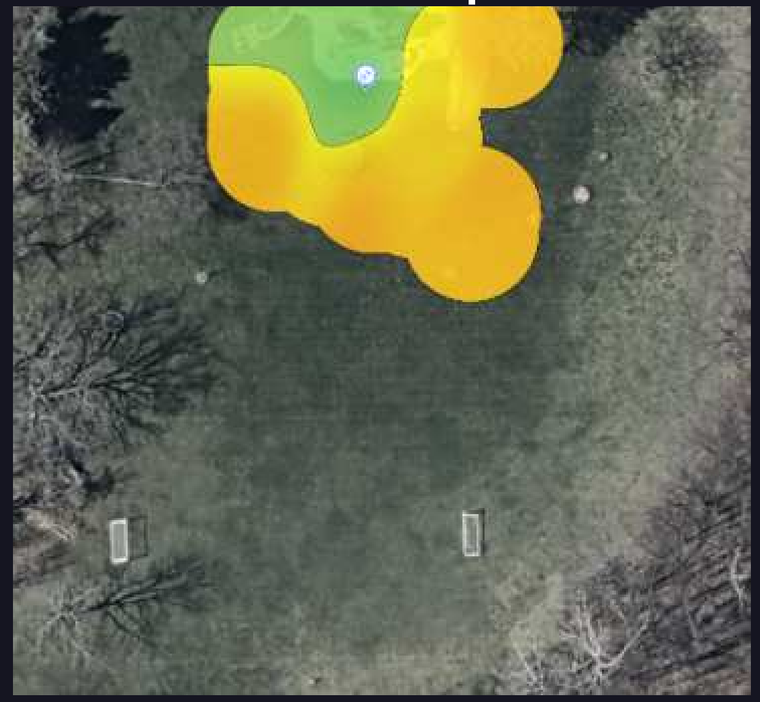
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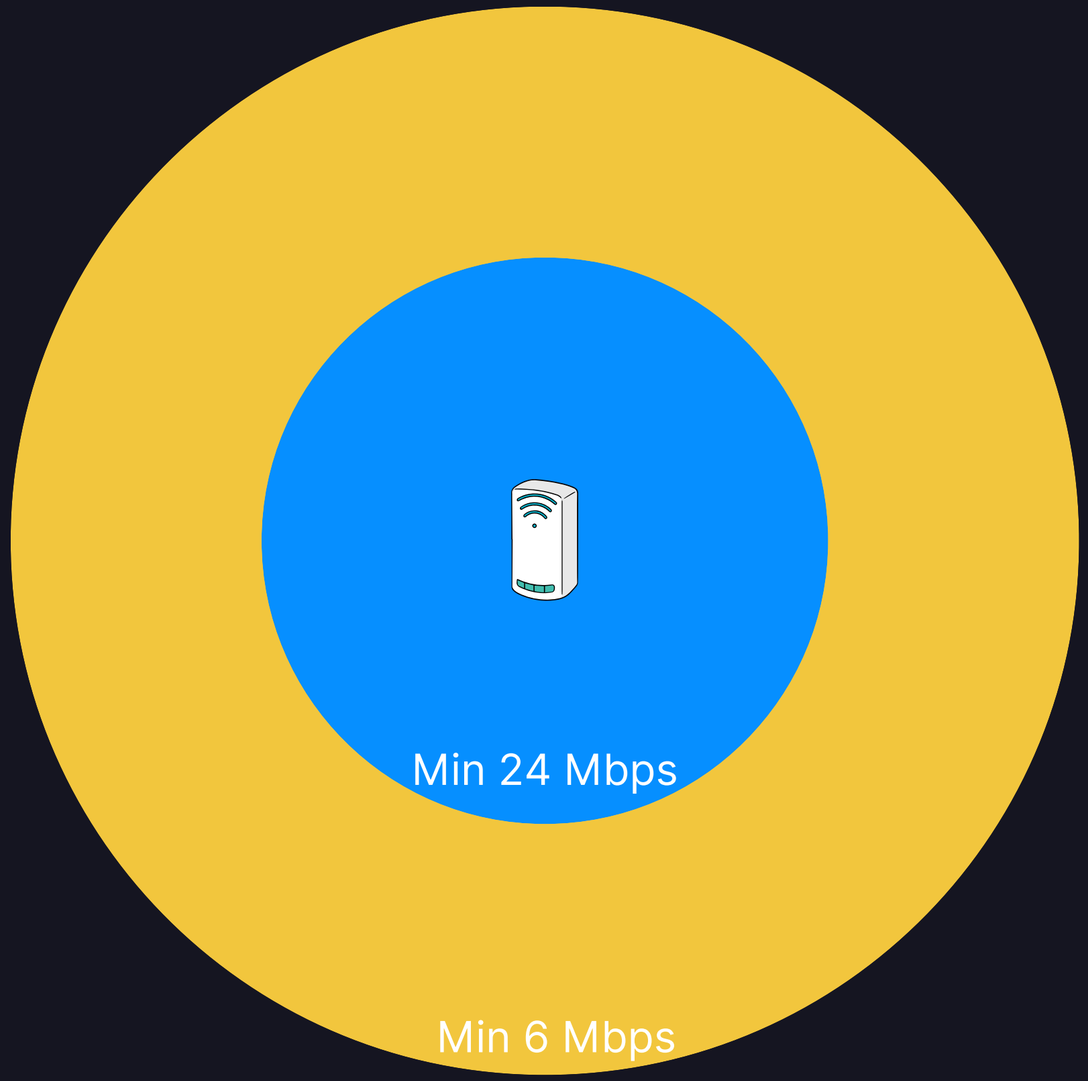
The environment didn’t change. The decode boundary did!

RF energy doesn’t vanish. Decodability does!!!

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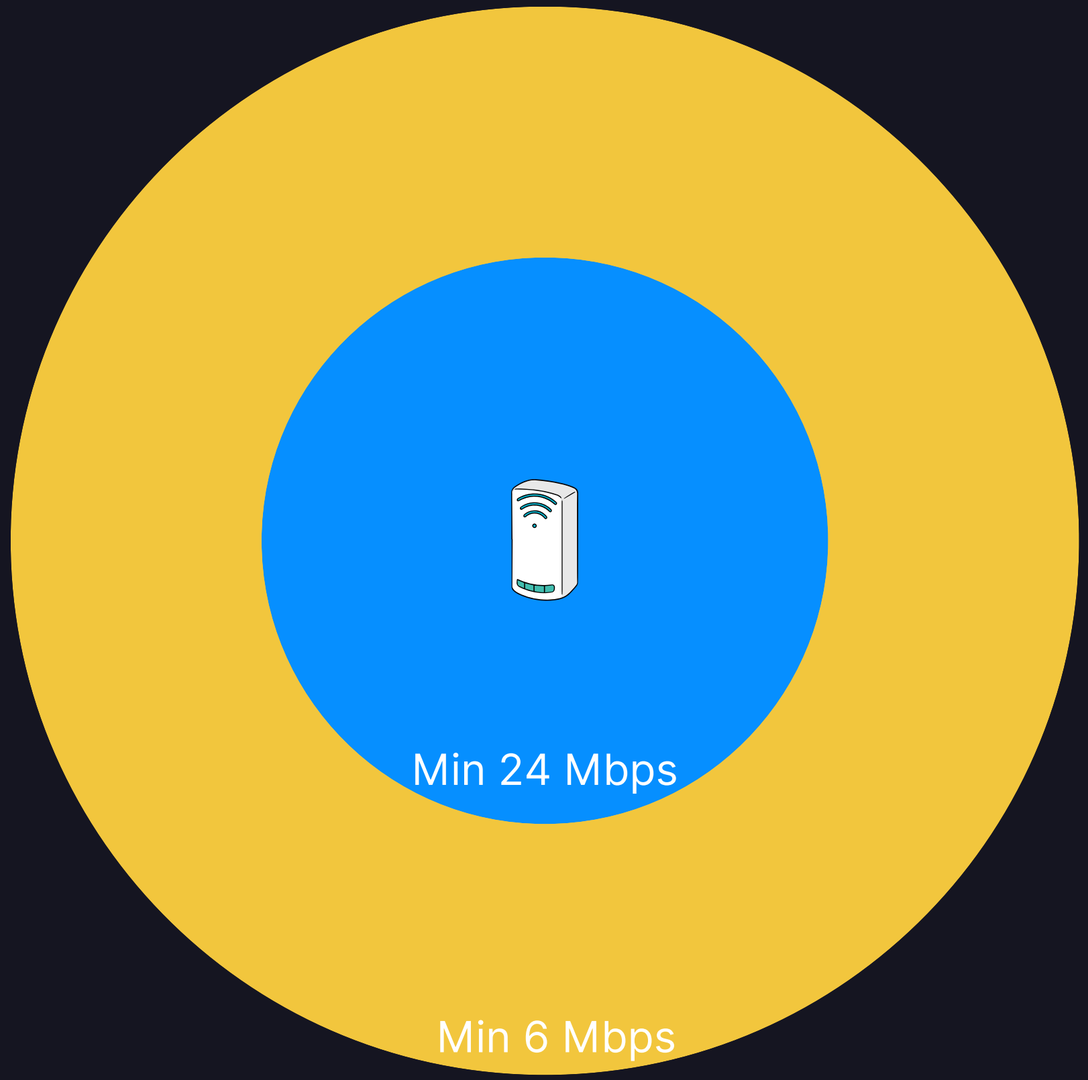

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At what number do you set your MBR?

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**Beacons use basic rates**

AP always sends beacons at the lowest allowed basic rate. Raising min data rate raises the beacon rate.




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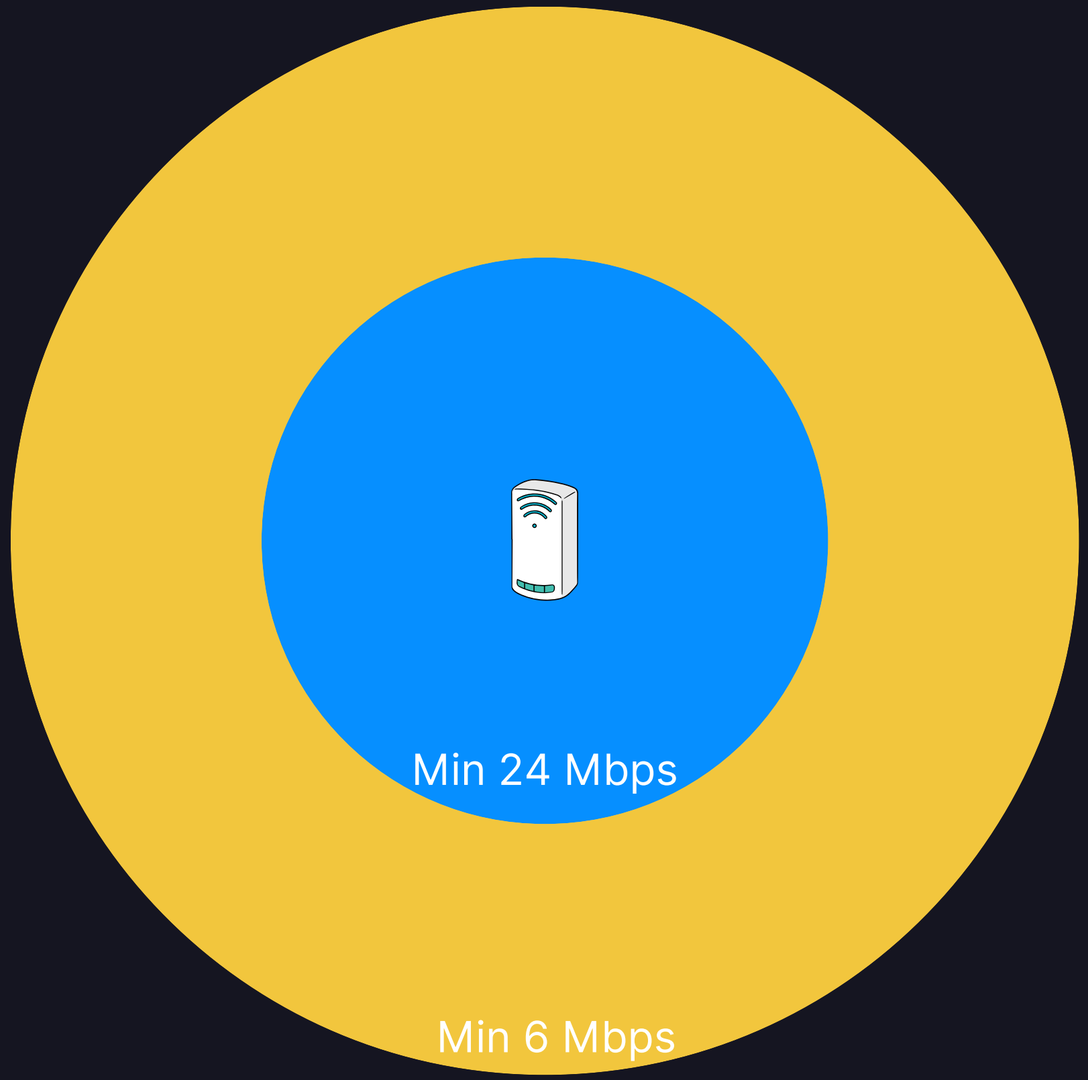
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**Higher rate = higher decode threshold**

Faster rates need a better SNR to decode. The receive threshold shifts; closer to the AP.




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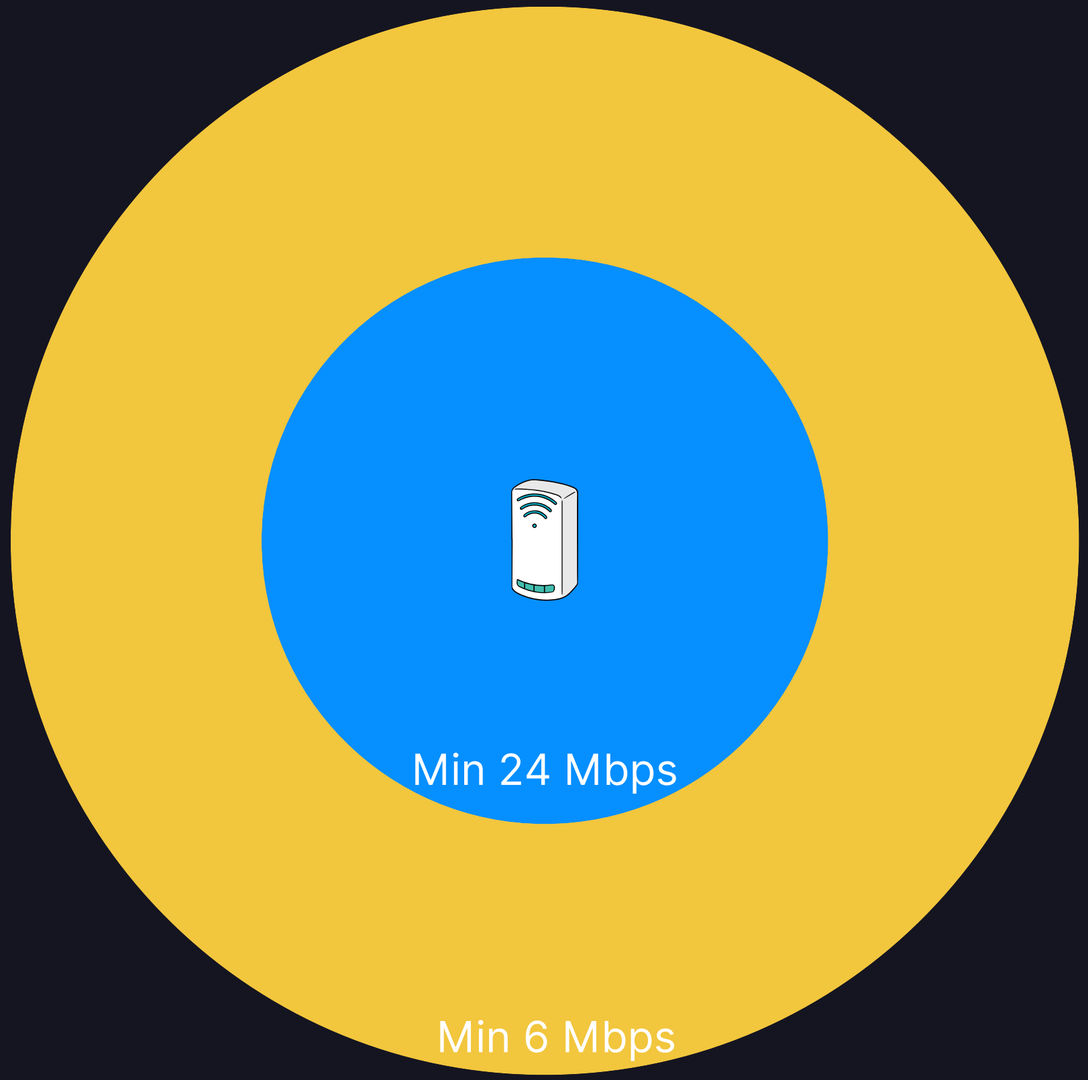
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**Frame becomes undecodable at edges**

At the boundary, signal drops below the decode threshold. The RF is still there; it just can’t be decoded.

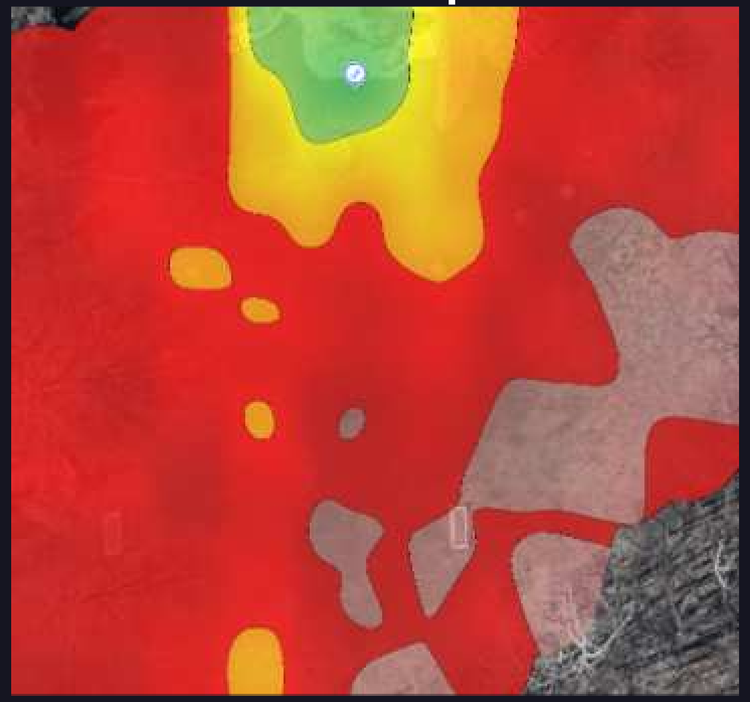



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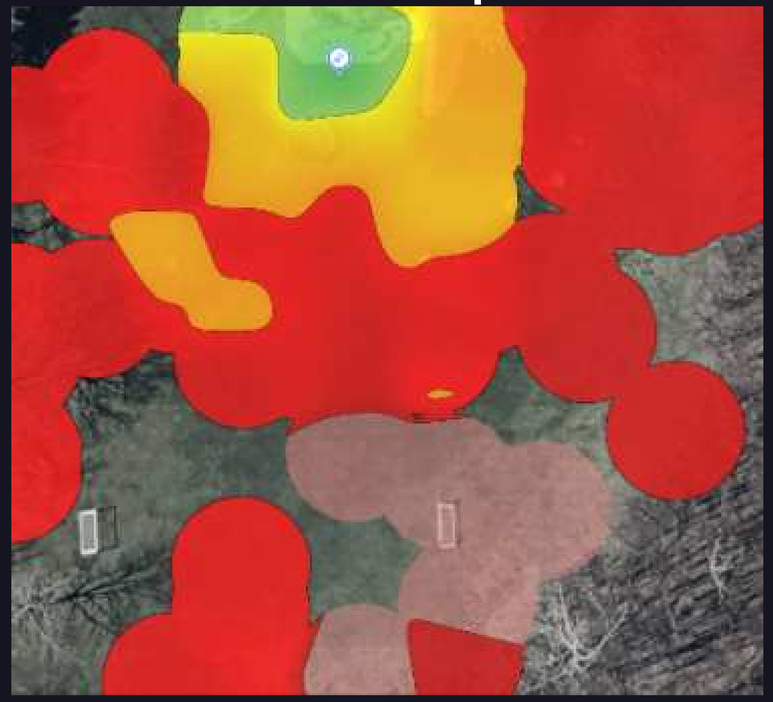
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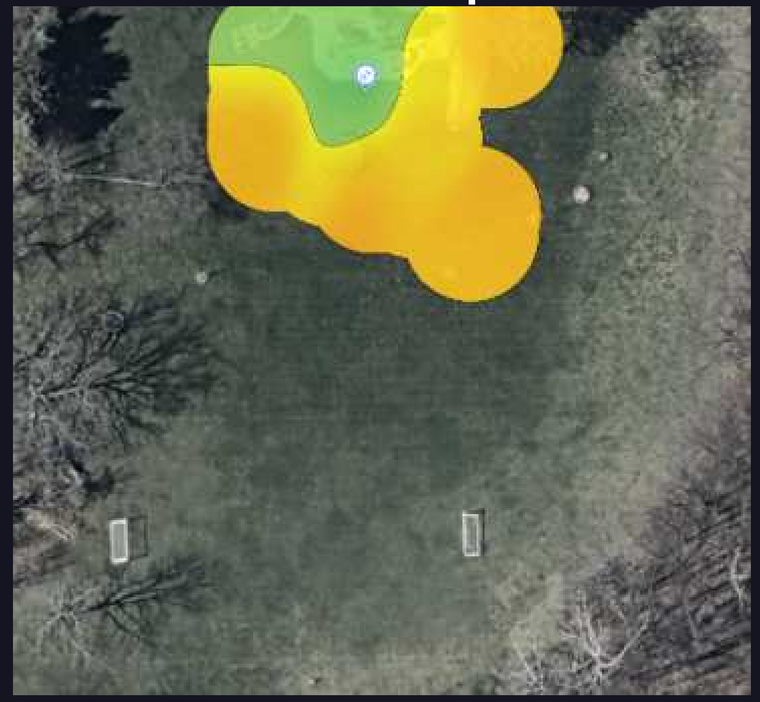
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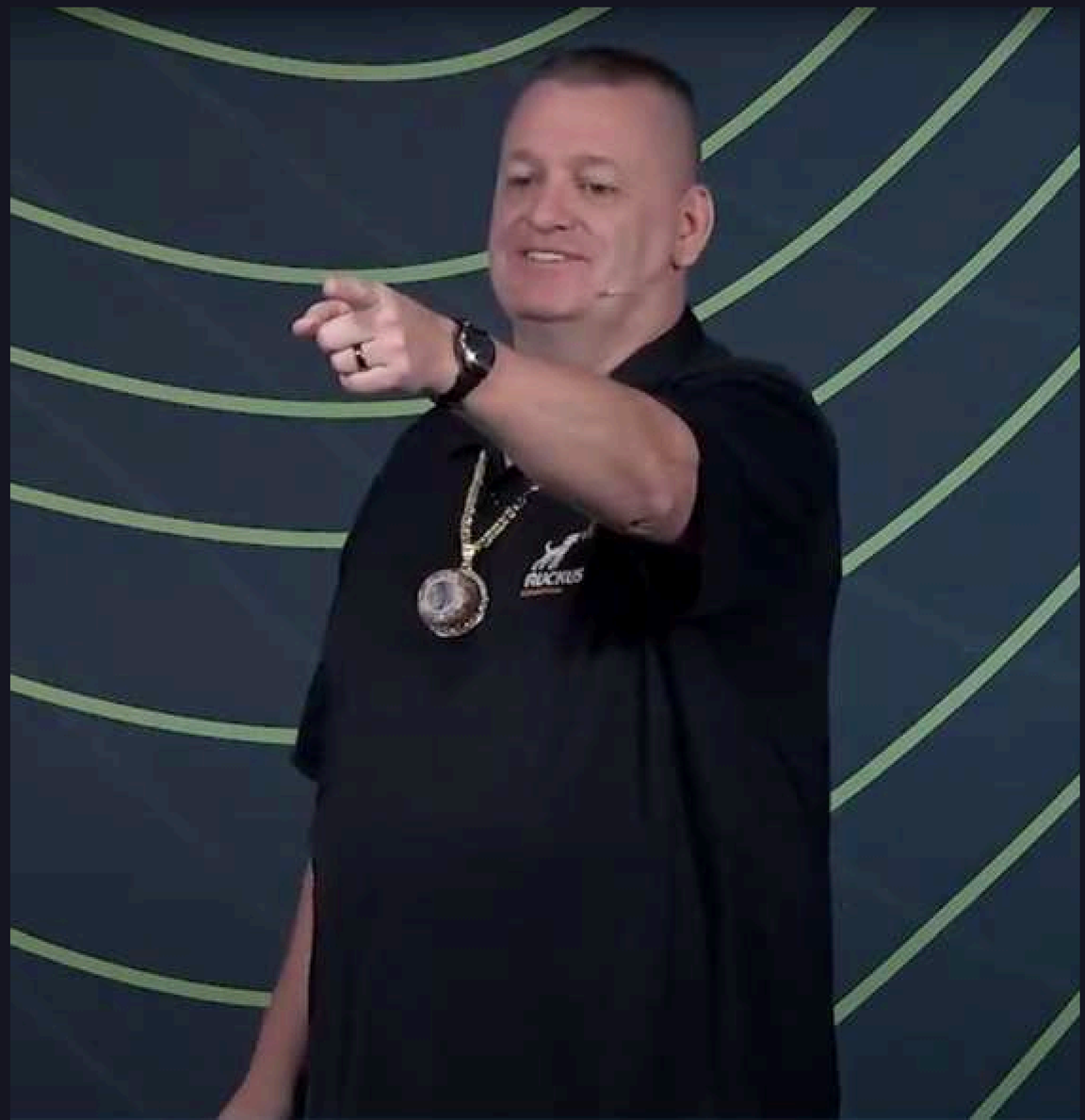
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# “The dumbest member of the network, is the one who decides for the network!”

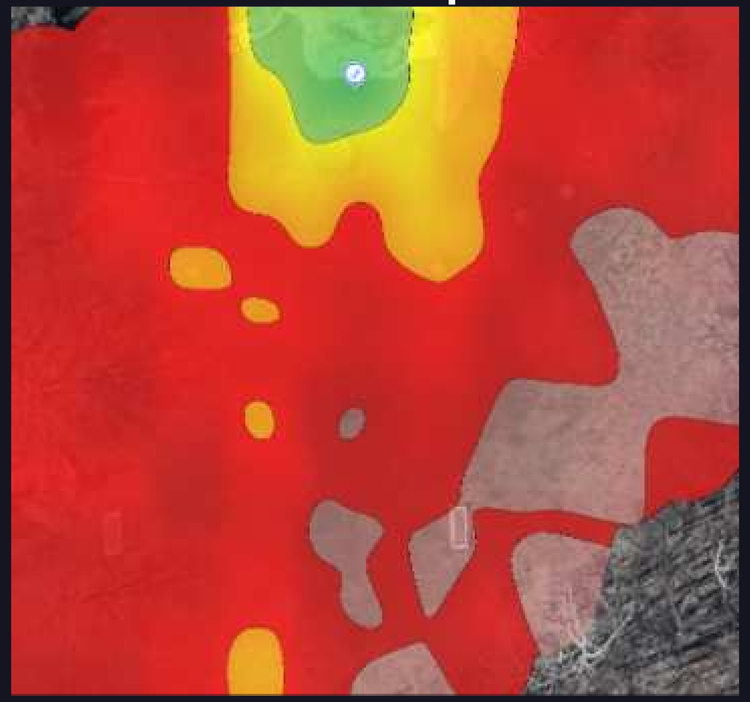


Jim Palmer
Principal Technical Evangelist at RUCKUS Networks

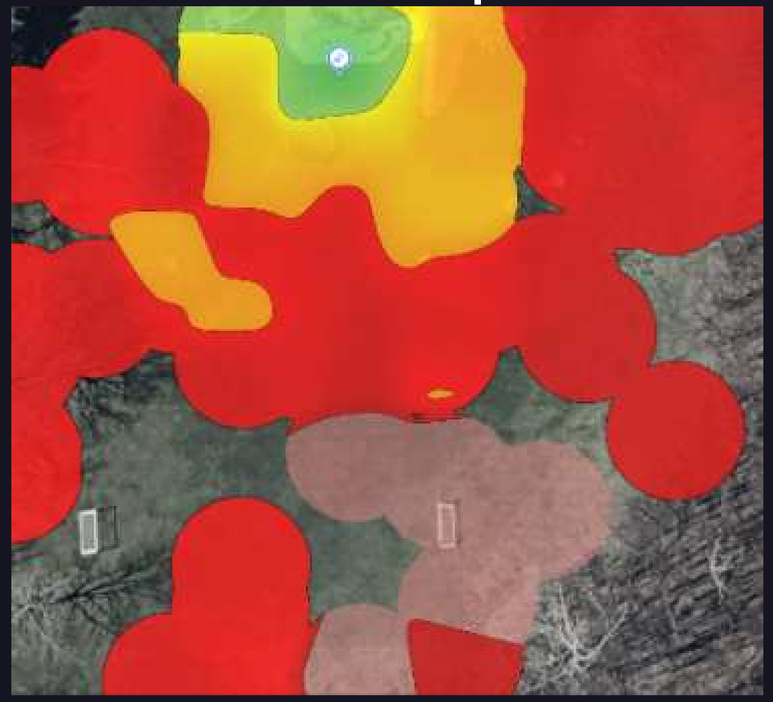
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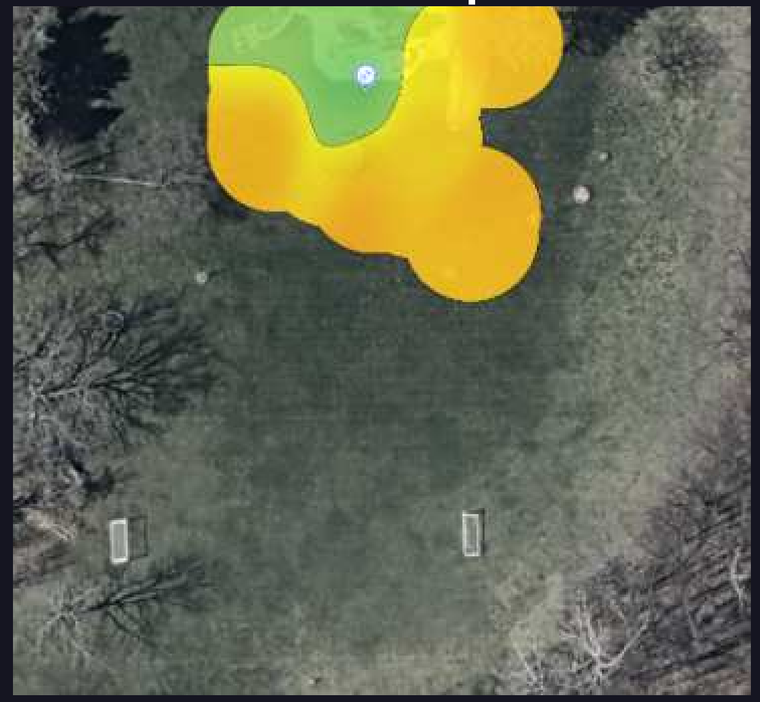
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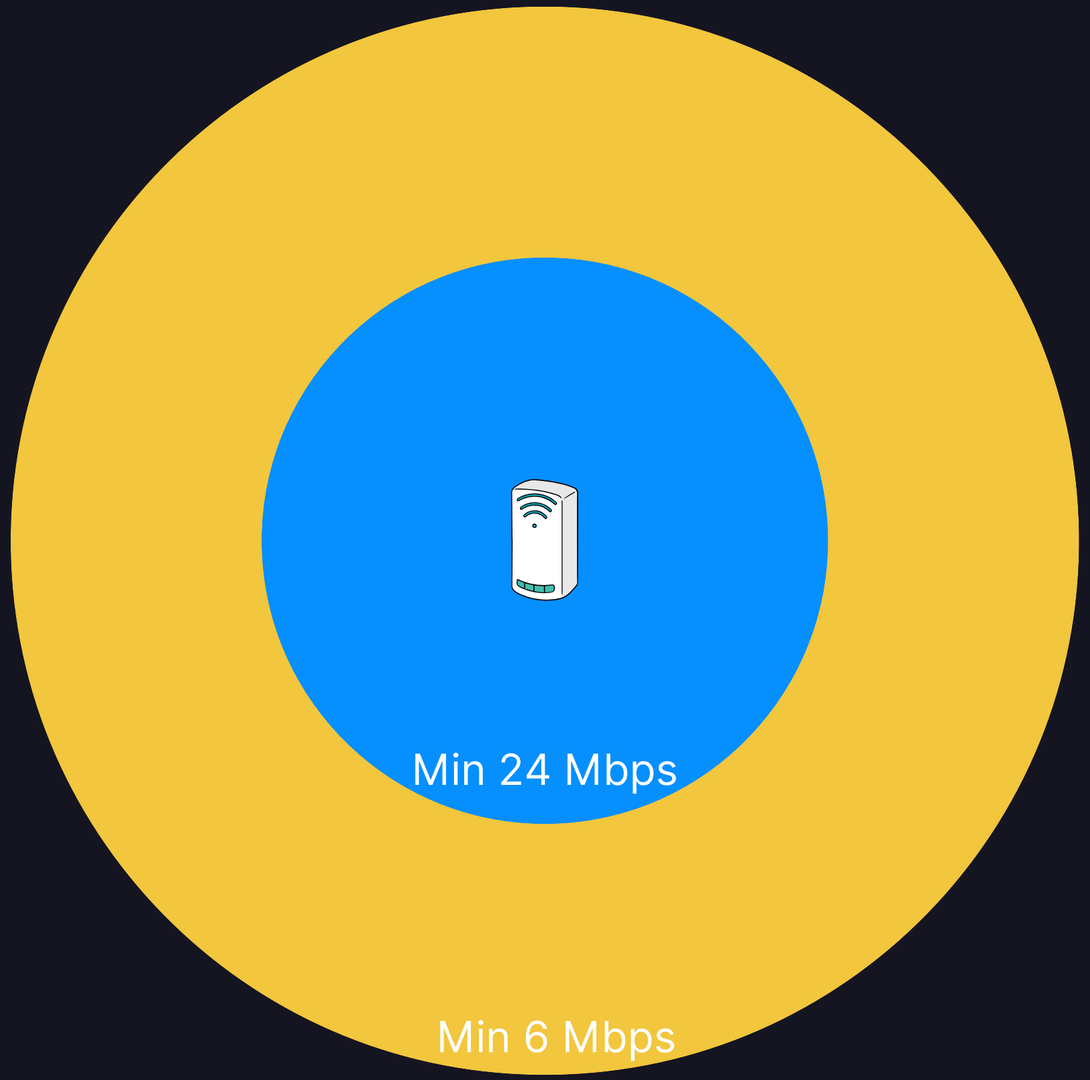
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These BSA circles are based on beacons!

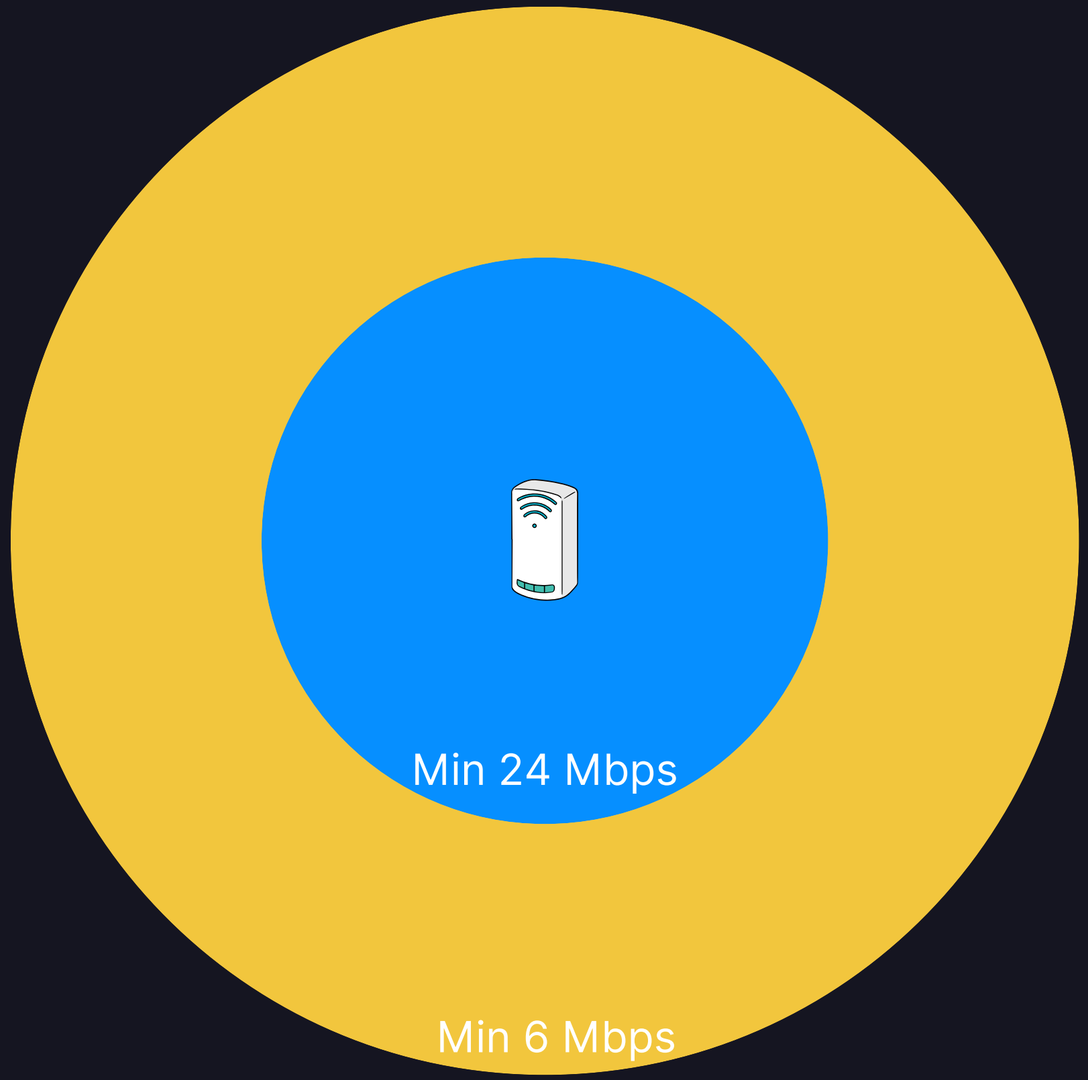



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**Beacon view**
Fixed, low rate




Data provided by: Jerry Olla

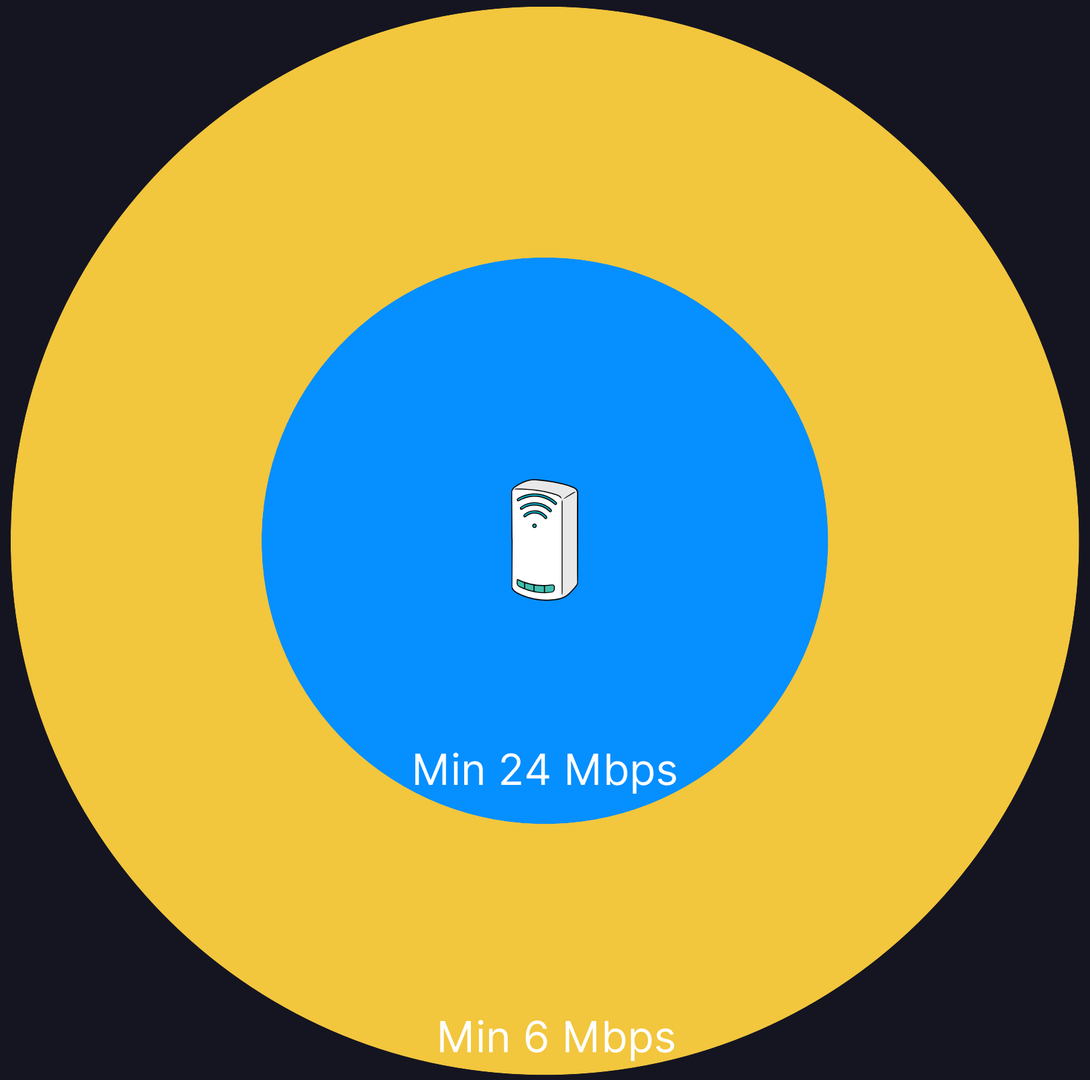
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**Beacon view**
Fixed, low rate

**Data Frame (not beacon!)**
Adaptive MCS (rate selection)




Data provided by: Jerry Olla

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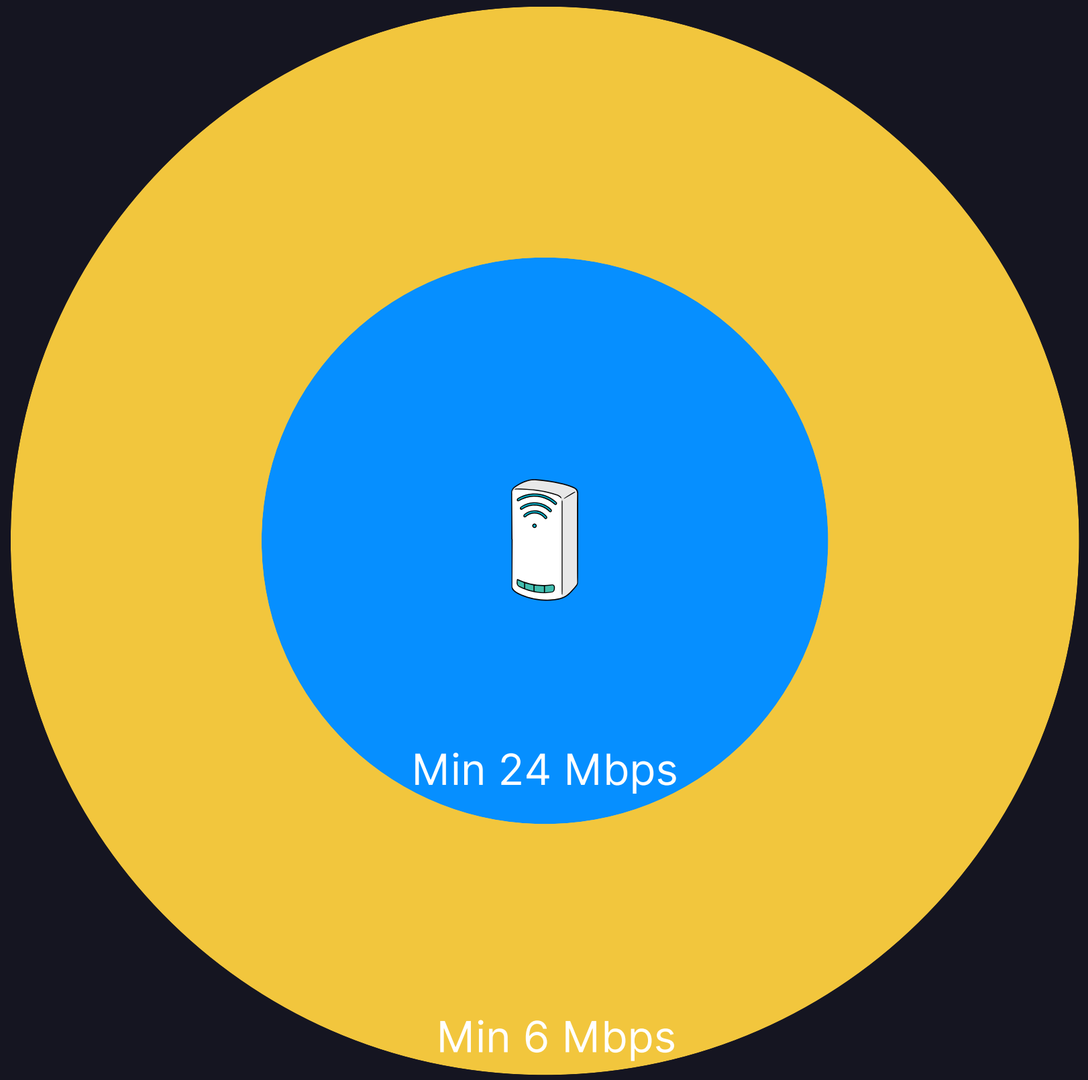
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**Beacon view**
Fixed, low rate

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**User Experience**
What we care about!




Data provided by: Jerry Olla

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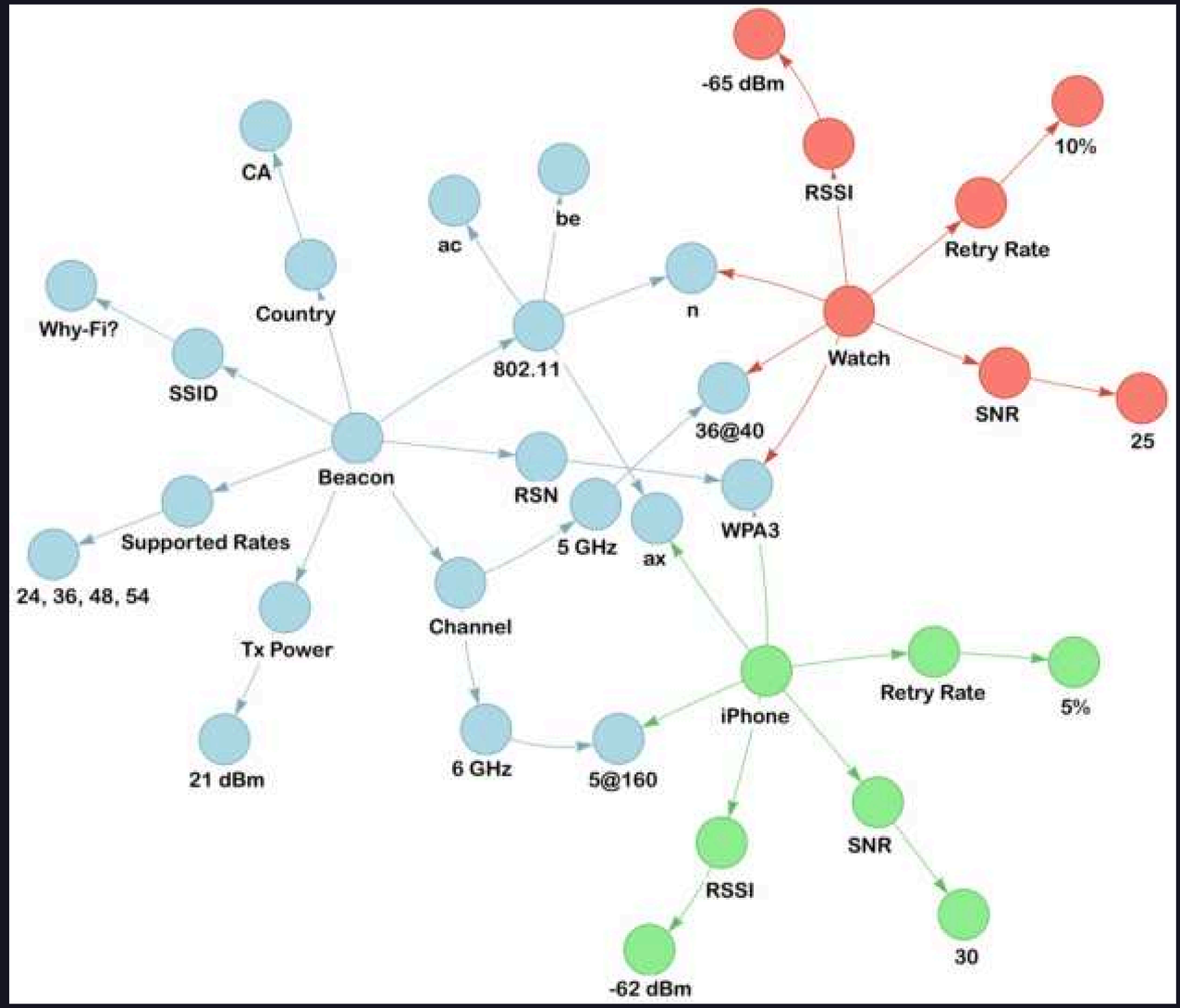
# What's My Client's Wi-Fi Experience?



# What piece of information do I need?

The client side:

These tie to the client's specifications and capabilities.



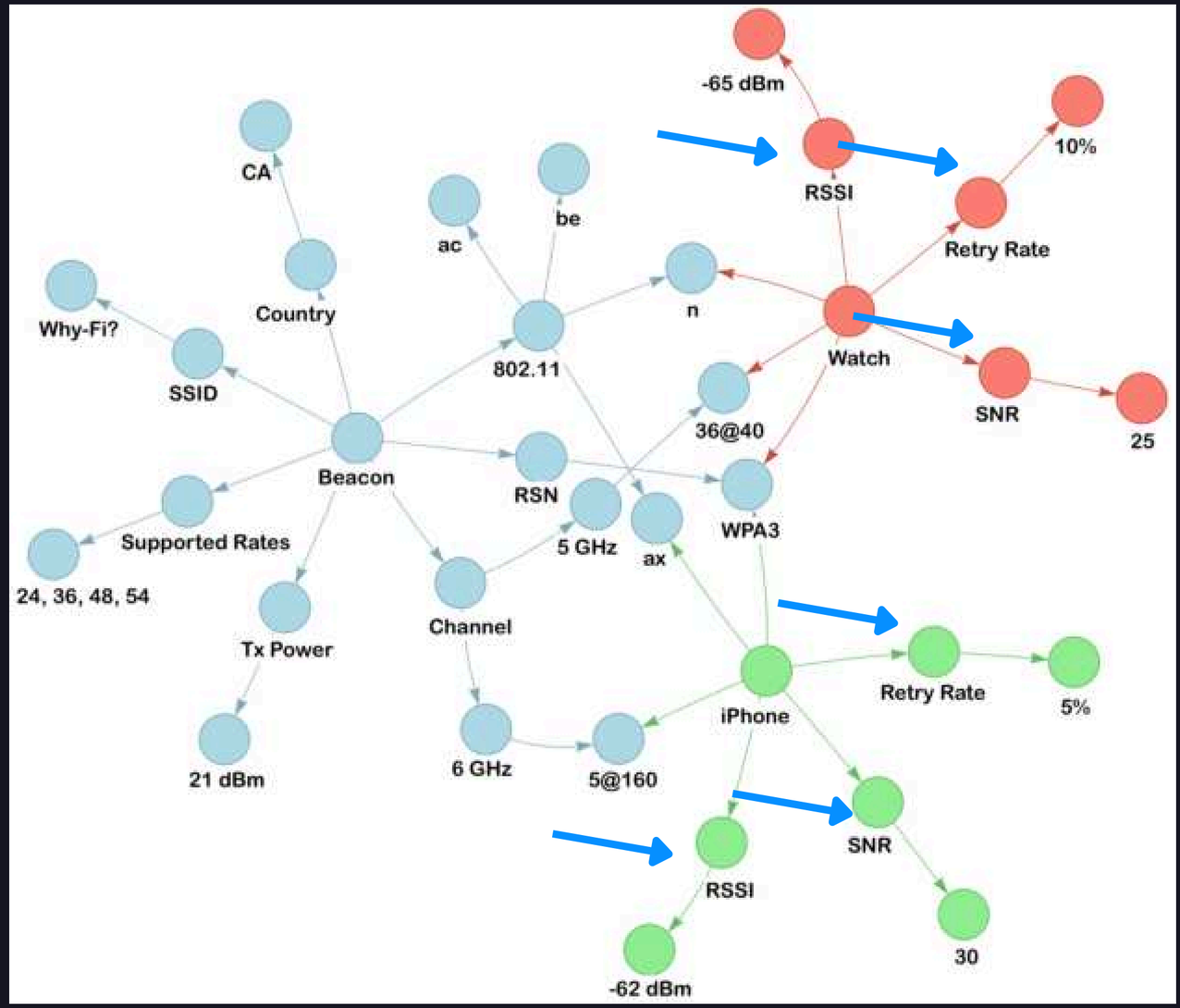
# What piece of information do I need?

The client side:

These tie to the client's specifications and capabilities.

What can affect them?

- FSPL
- Noise
- Interference
- Density
- Environment Attenuation
- Receiver Sensitivity
- Battery Level
- Device Orientation



# Device Orientation

Same AP.

Same moment.

Different orientation.

Data provided by: Jerry Olla

# Device Orientation

Same AP.

Same moment.

Different orientation.

The way someone holds their device changes RSSI



Data provided by: Jerry Olla

# Device Orientation

Same AP.
Same moment.
Different orientation.

| Orientation | RSSI | Δ |
|---|---|---|
| **Front (facing AP)** | **–60 dBm** | *baseline* |
| **Side (90°)** | **–66 dBm** | **6 dB** |
| **Back (facing away)** | **–71 dBm** | **11 dB** |

The way someone holds their device changes RSSI

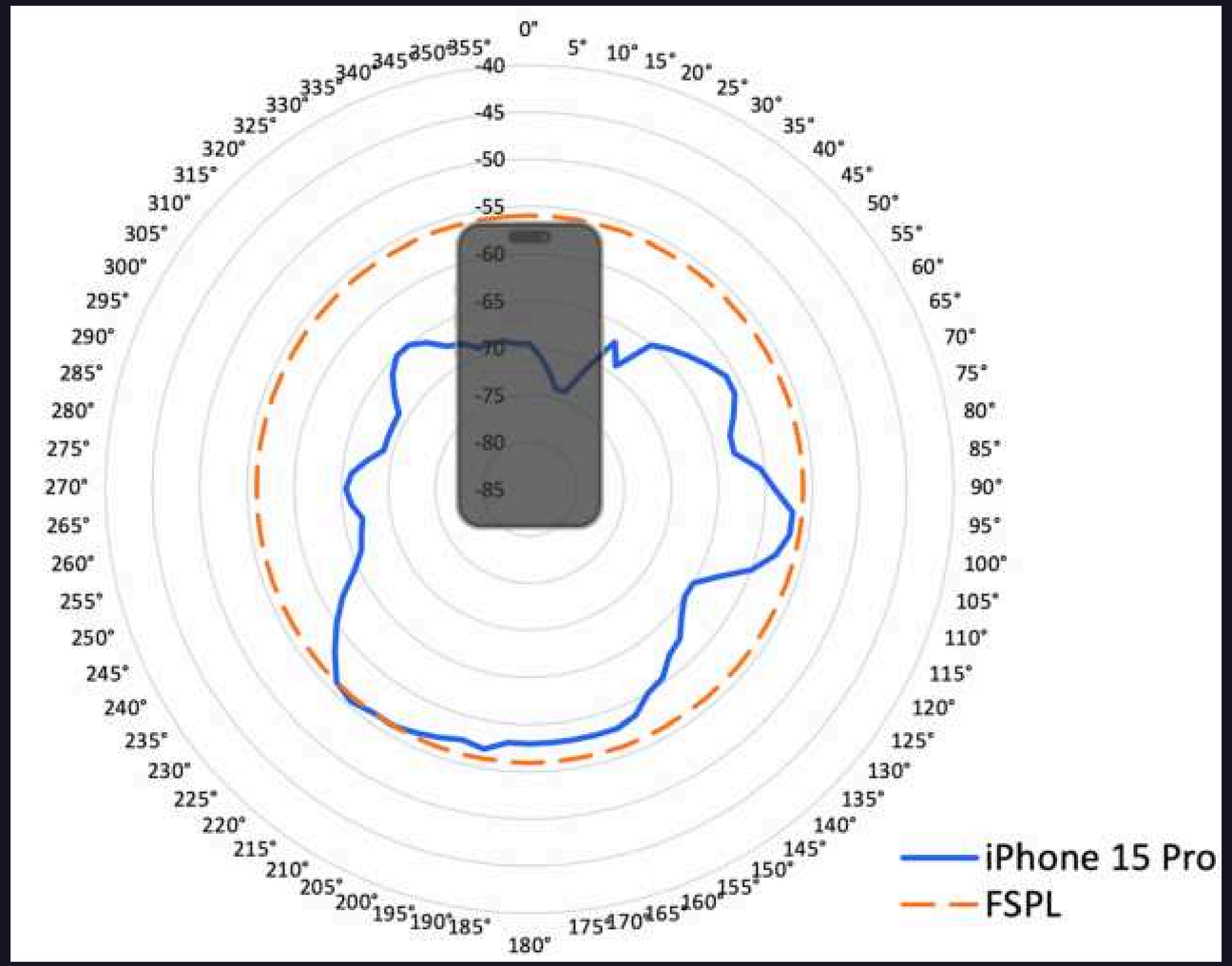


An 11 dB swing, from the same device,
in the same spot.

Data provided by: Jerry Olla

# Device Orientation

## iPhone 15 Pro




## Device Stats

**18 dB**
Signal Range

**−7.6 dB**
Avg Deviation

**−18 dB**
Worst Deviation

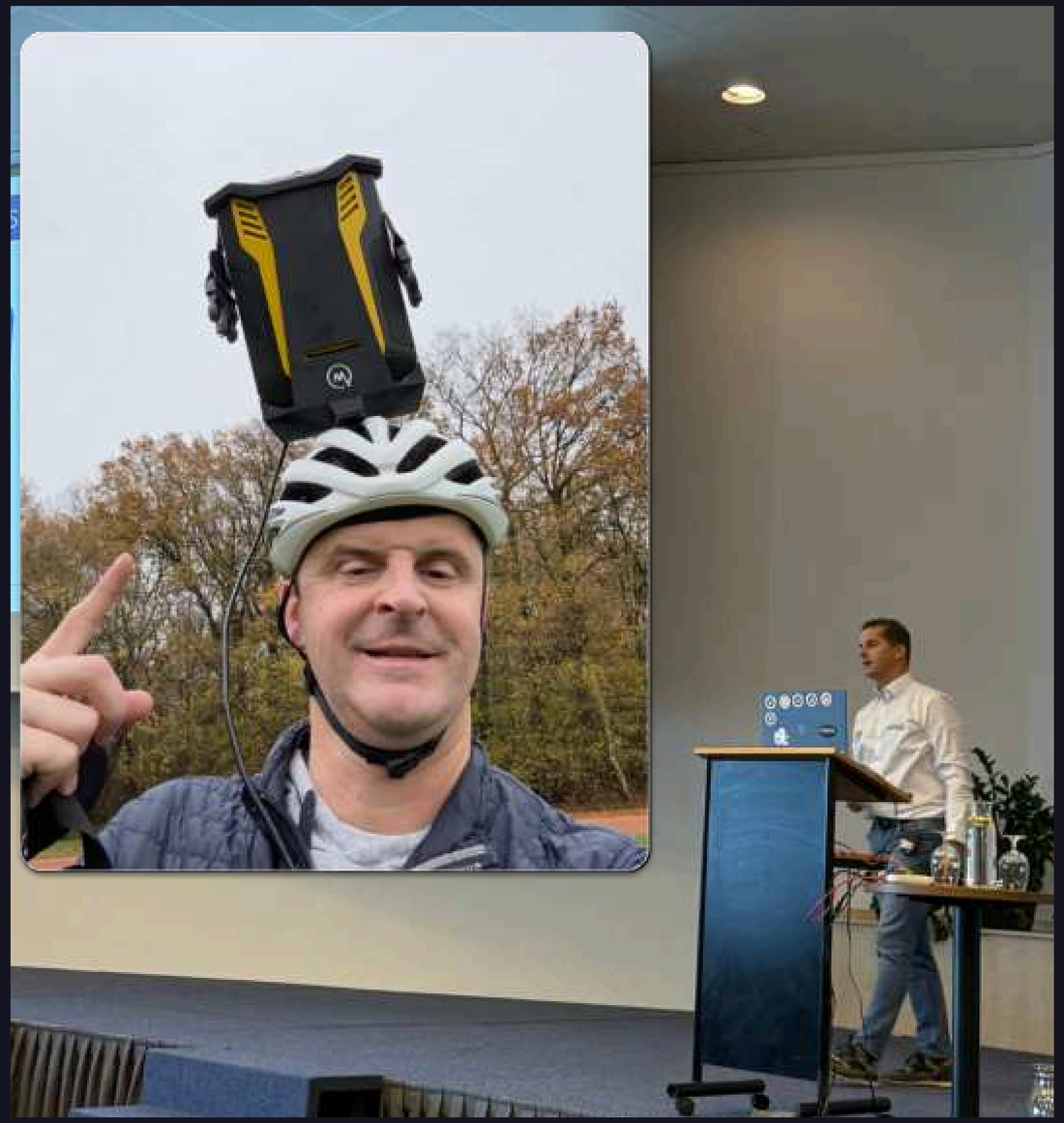
**0 dB**
Best Deviation

**±4.9 dB**
Std Deviation

Data provided by: Jerry Olla

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# Hans Van Ballaer

Owner of AHFIBIT BV

CWNE#340

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# What's My Client's Wi-Fi Experience?



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# What's My Client's Wi-Fi Experience?

To answer this:
I tried to create a model that analyzes a client's health by integrating metrics belonging to both Layers 1 and 2.



# Client’s Health

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I need to build a model that:

Assigns rewards and penalties to the received statistics.

(Reinforcement Learning)

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But, it’s not that simple! 🤔☹️

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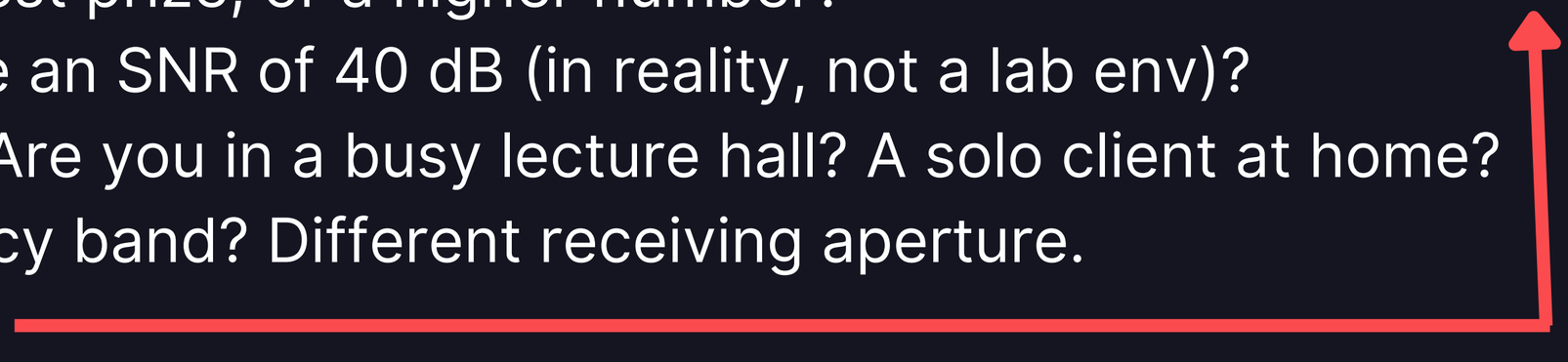
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6 GHz, 160 MHz
The wider the channel,
The higher the PSD!



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**“There will be a lot of trade-offs!
And I applied them based on my measurements.”**

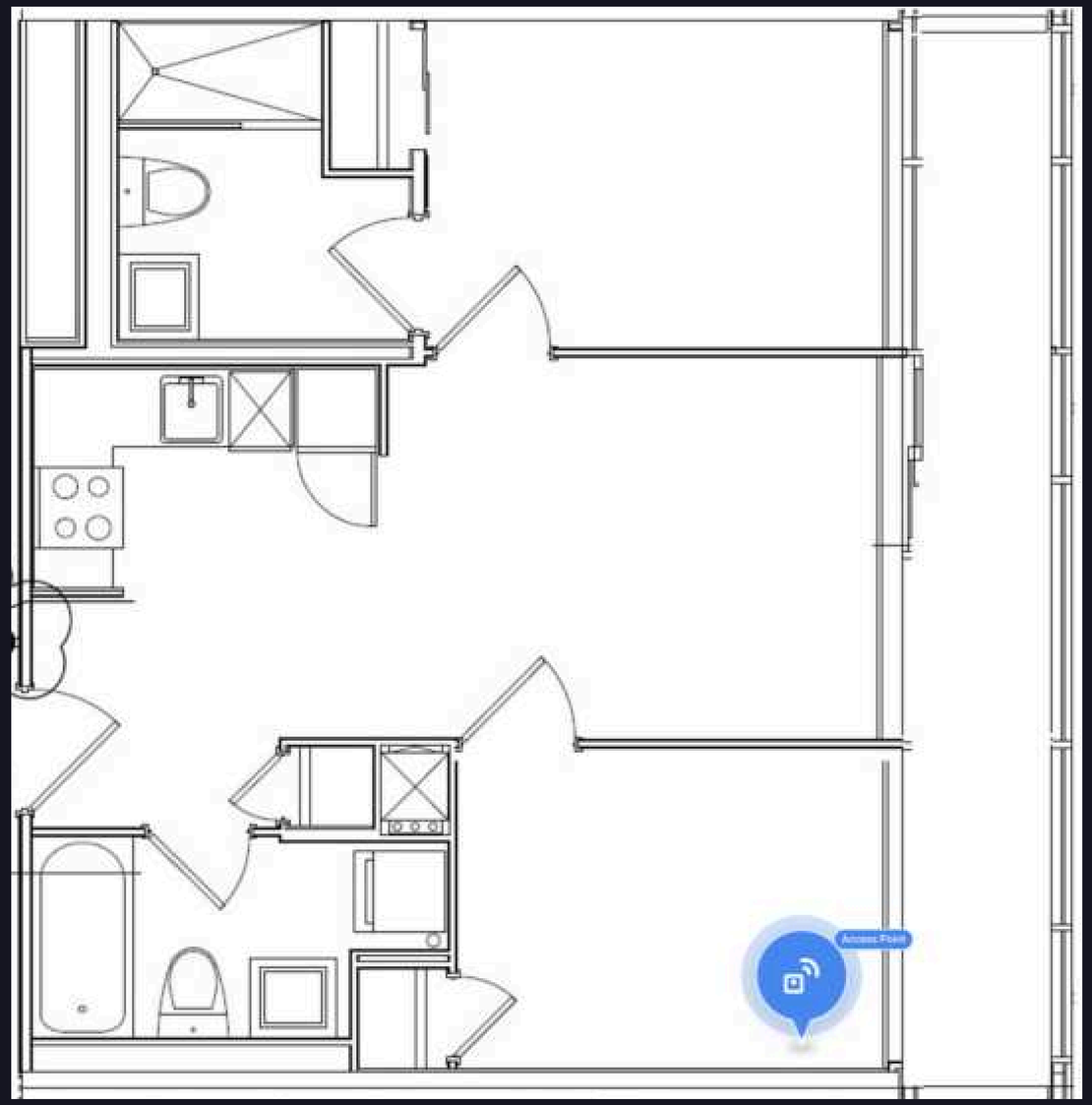
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# Client’s Health

Let’s set up the environment!



# Client’s Health

AP34, Wi-Fi 6E, 2 Spatial Streams

Mist AP is located in the bottom-right corner at 70 cm height, facing up.

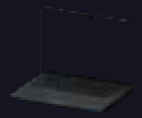
| SSID | Frequency | Channel Width | Tx Power | MBR |
|---|---|---|---|---|
| Why-Fi? | 5 GHz | 40 MHz | 18 dBm | 24 Mbps |
| Why-Fi? | 6 GHz | 160 MHz | 21 dBm | 24 Mbps |



# Client’s Health


AP34, Wi-Fi 6E, 2 Spatial Streams

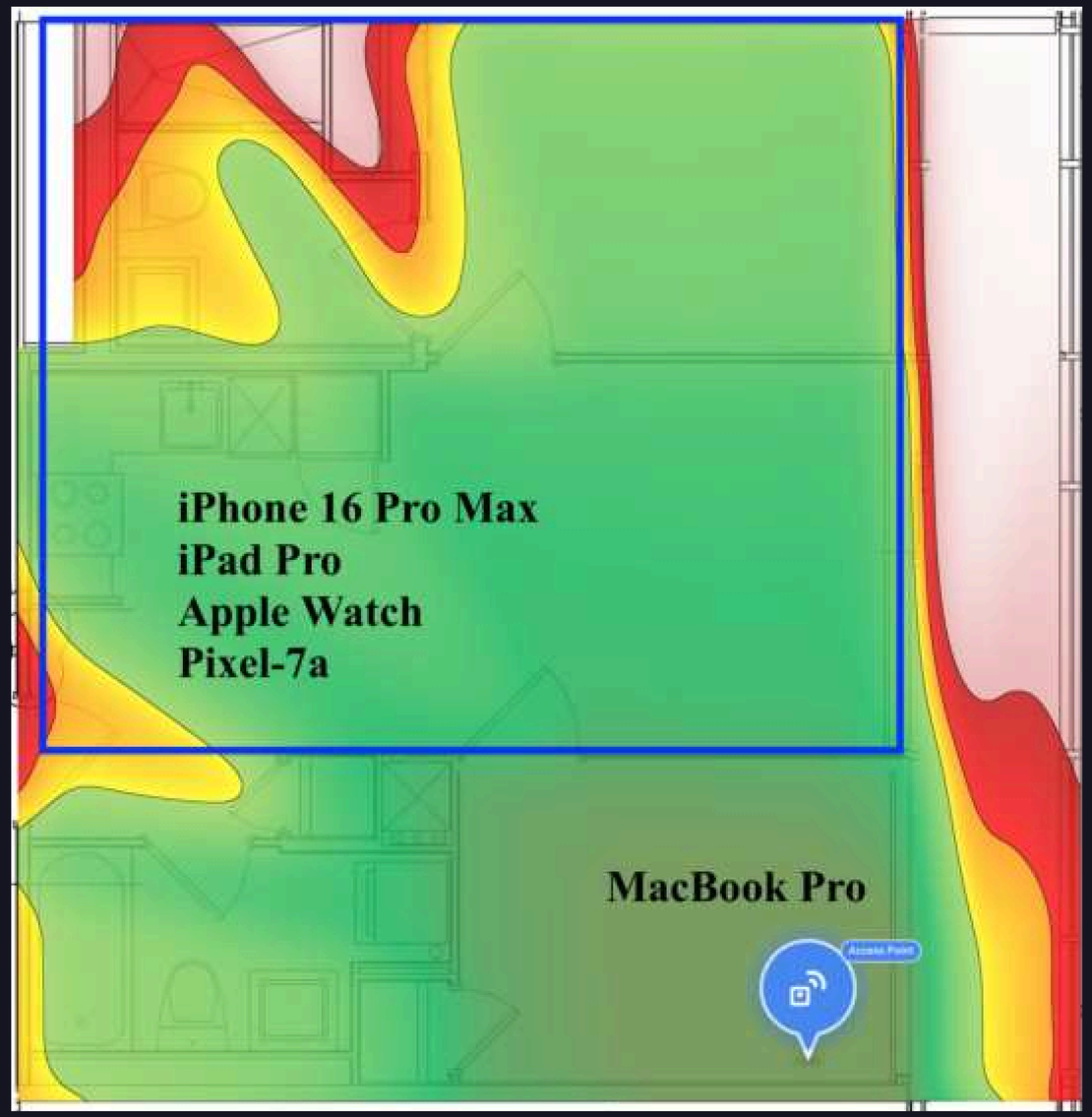

iPhone 16 Pro Max, Wi-Fi 7, 2 Spatial Streams


MacBook Pro, Wi-Fi 6E, 2 Spatial Streams


iPad Pro, Wi-Fi 6, 2 Spatial Streams


Google Pixel 7a, Wi-Fi 6E, 2 Spatial Streams


Apple Watch Series 9, Wi-Fi 4, 1 Spatial Stream



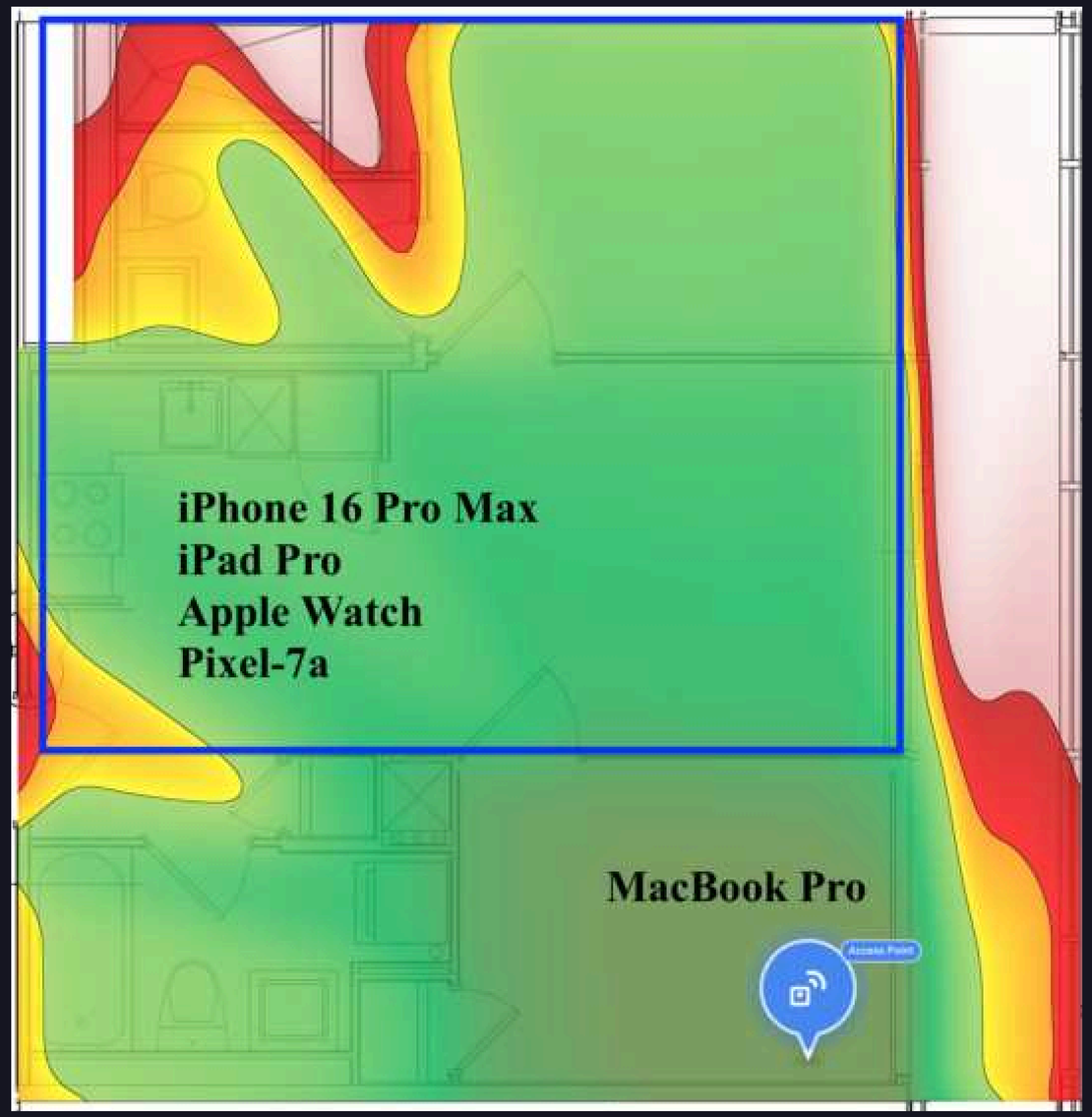
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# Client's Health

Python Script ⇄ HPE JUNIPER Mist

- AP34, Wi-Fi 6E, 2 Spatial Streams
- iPhone 16 Pro Max, Wi-Fi 7, 2 Spatial Streams
- MacBook Pro, Wi-Fi 6E, 2 Spatial Streams
- iPad Pro, Wi-Fi 6, 2 Spatial Streams
- Google Pixel 7a, Wi-Fi 6E, 2 Spatial Streams
- Apple Watch Series 9, Wi-Fi 4, 1 Spatial Stream



# Client’s Health




RSSI & SNR (layer1):  From the AP NIC

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Retry Rate (layer2): [(number of retransmitted frames) / (number of all data frames)] x 100

# Client’s Health



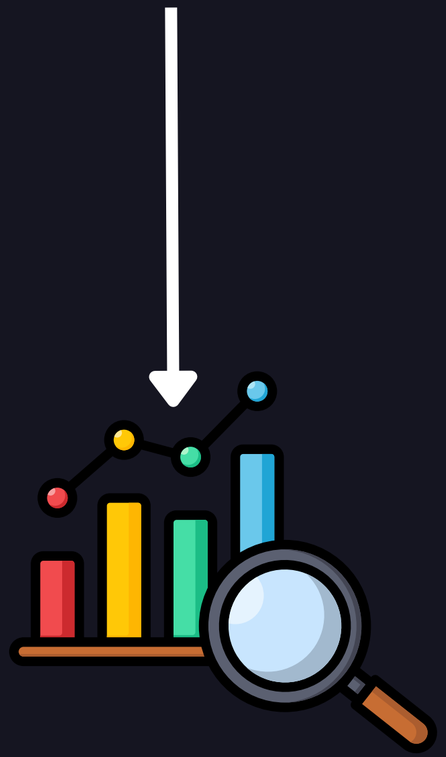

RSSI & SNR (layer1):  From the AP NIC

Retry Rate (layer2): [(number of retransmitted frames) / (number of all data frames)] x 100

MCS Index: To achieve this the model should reverse engineer the MCS Index tables based on the client’s data rates.

# Client’s Health

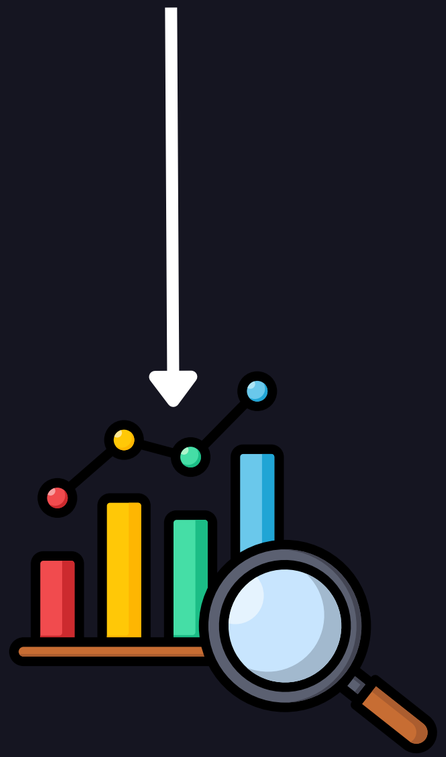


Four different clients were monitored over a two-hour recording window. This test was carried out more than 10 times to refine the scoring system and model.

# Client’s Health



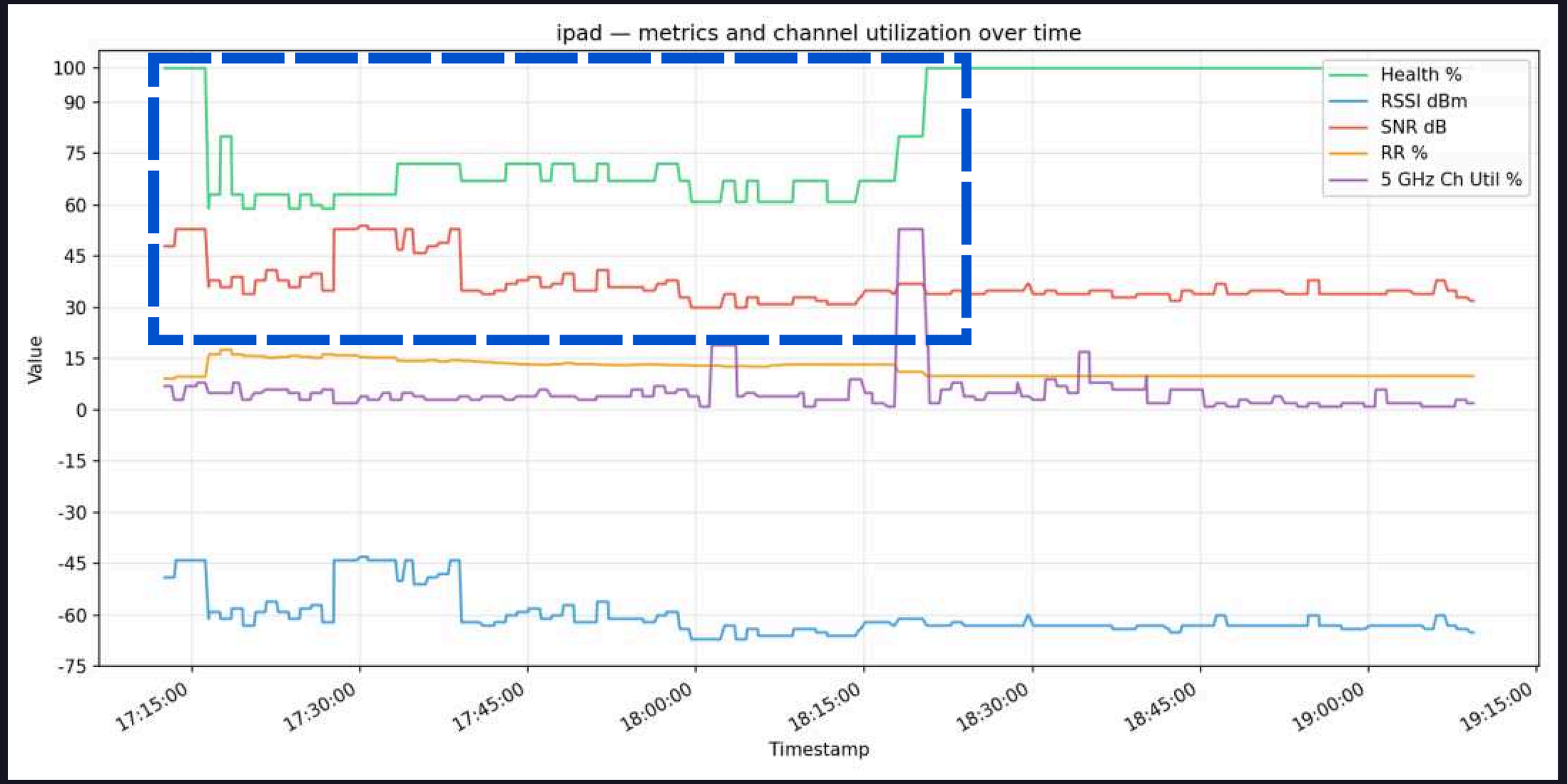
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Let’s see the results.

# iPad Pro 3rd Generation 

- Was used to watch a show on Disney+
- No lag, no glitch, it performed without a blink.

# iPad Pro 3rd Generation



# iPad Pro 3rd Generation



# My Learning 1:

A user (Layer7) may still be satisfied with the service while the device (Layer 1 & 2) is struggling to maintain the connection, depending on the application (and its tolerance to delay, loss, or rate changes).

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Successful playback on streaming platforms such as Disney+, Netflix, or YouTube does not necessarily indicate that the wireless network health is “100%” on L1 & L2.

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# My Learning 2:

Successful playback on streaming platforms such as Disney+, Netflix, or YouTube does not necessarily indicate that the wireless network health is “100%” on L1 & L2.

# My Learning 3:

Good RSSI and SNR alone won’t guarantee good client health.

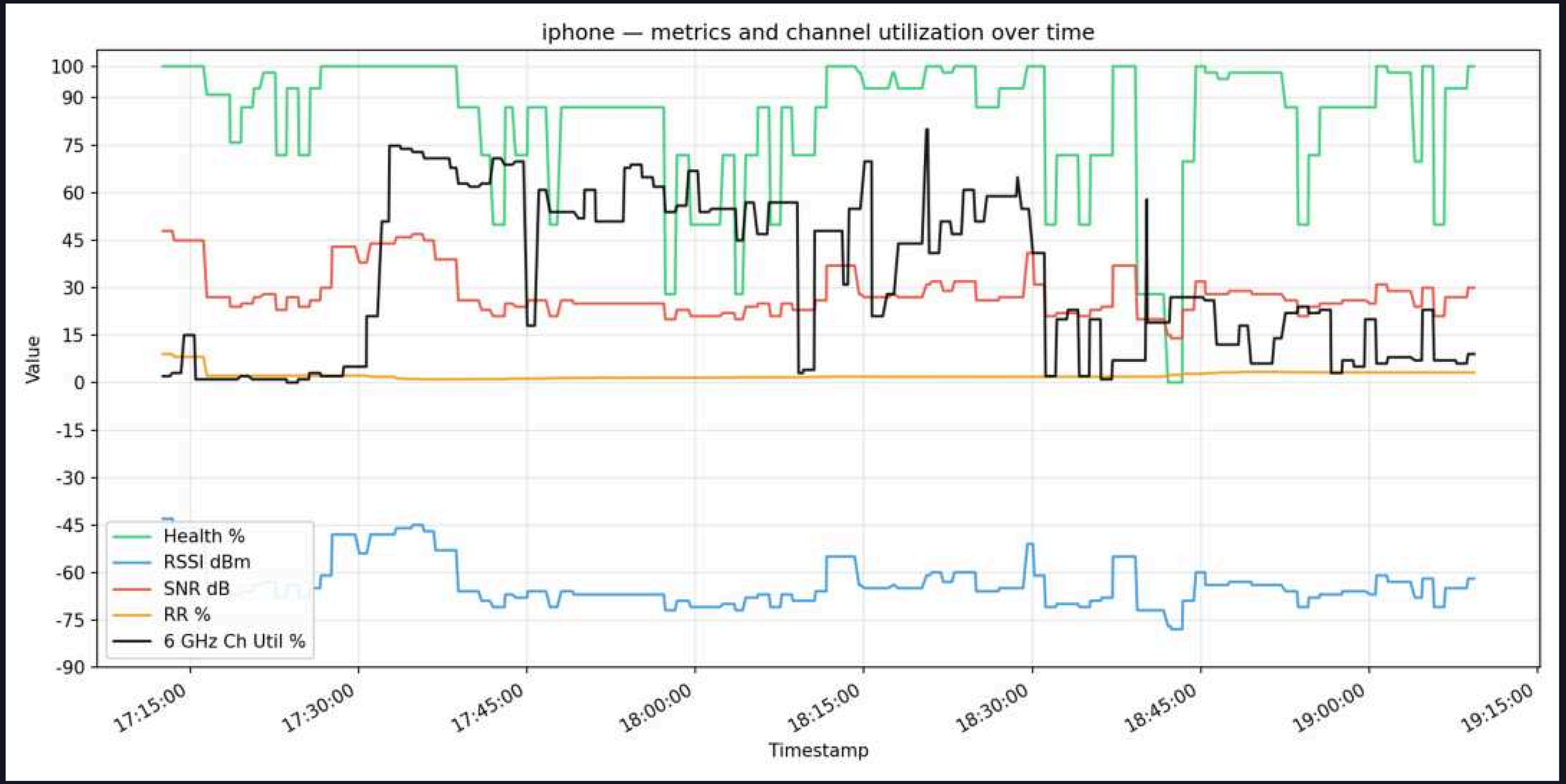
# iPhone 16 Pro Max



- Was used to have a WhatsApp video call + back-to-back nPerf tests.
- A Google Pixel 7a sat next to it, running speed tests.
- The device was moving around.

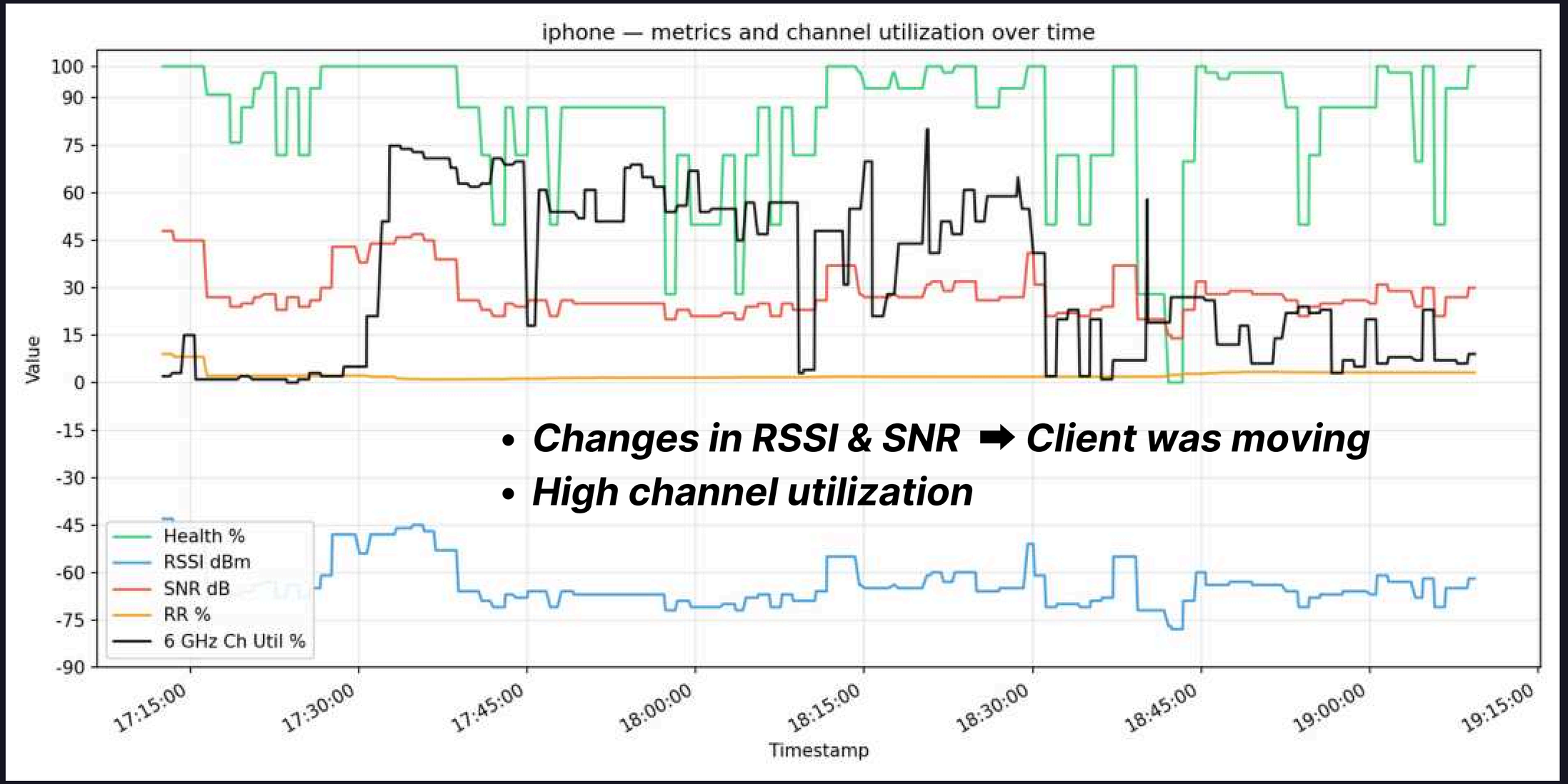
# iPhone 16 Pro Max





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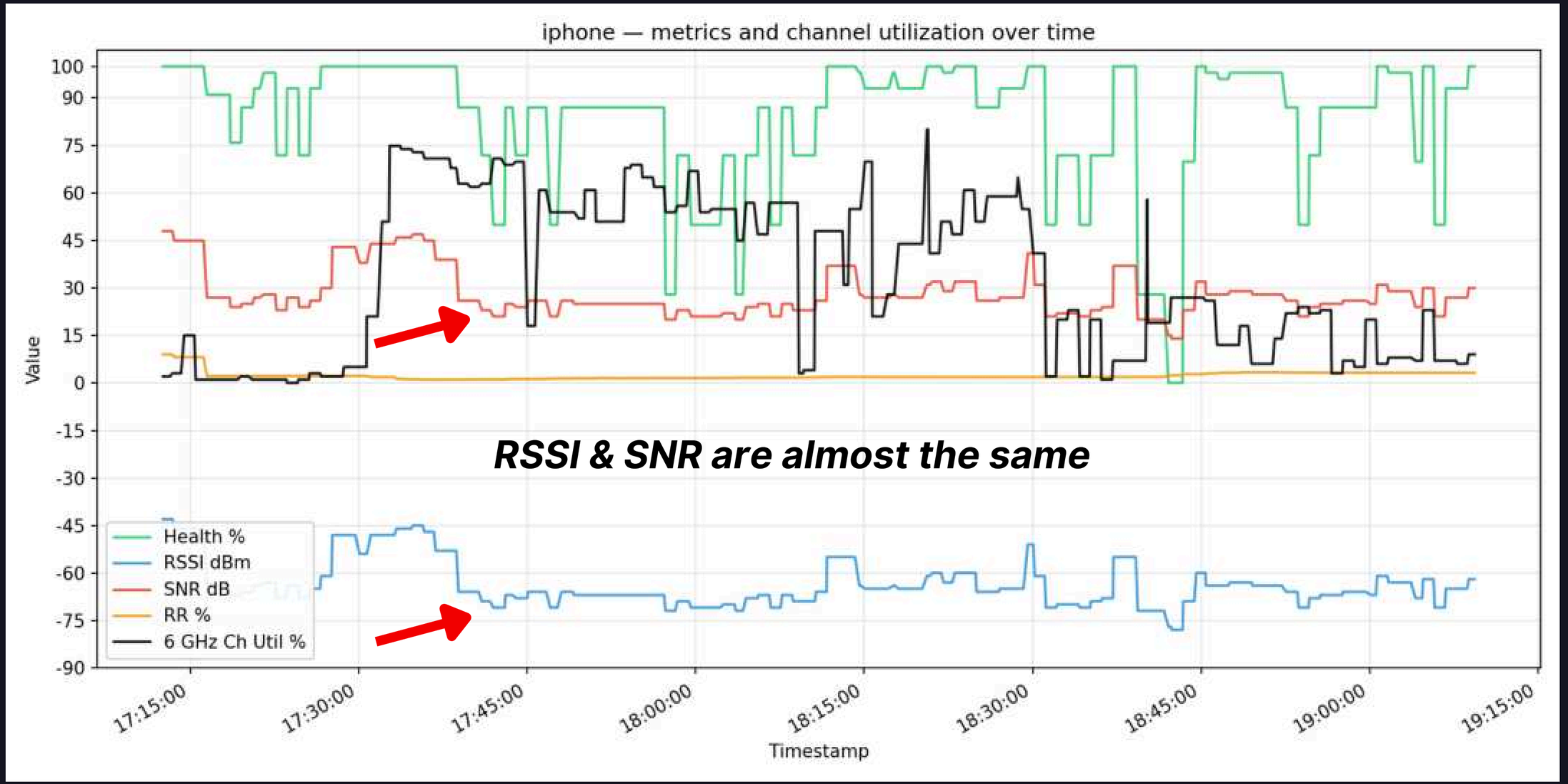





- ***Changes in RSSI & SNR ➡ Client was moving***
- ***High channel utilization***

# iPhone 16 Pro Max





# iPhone 16 Pro Max

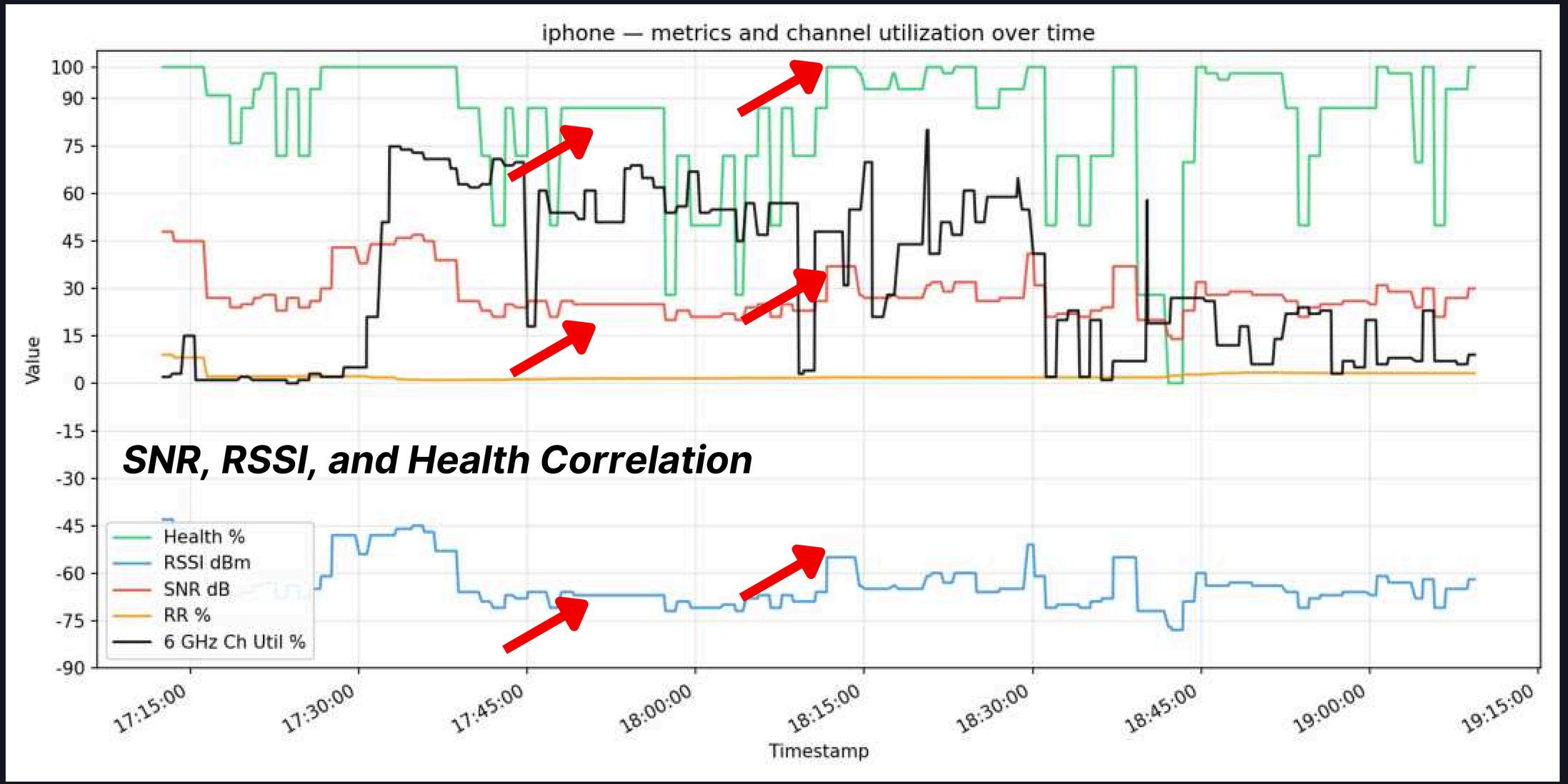


# My Learning 4:

There is a strong correlation between RSSI and SNR.

# iPhone 16 Pro Max






***SNR, RSSI, and Health Correlation***

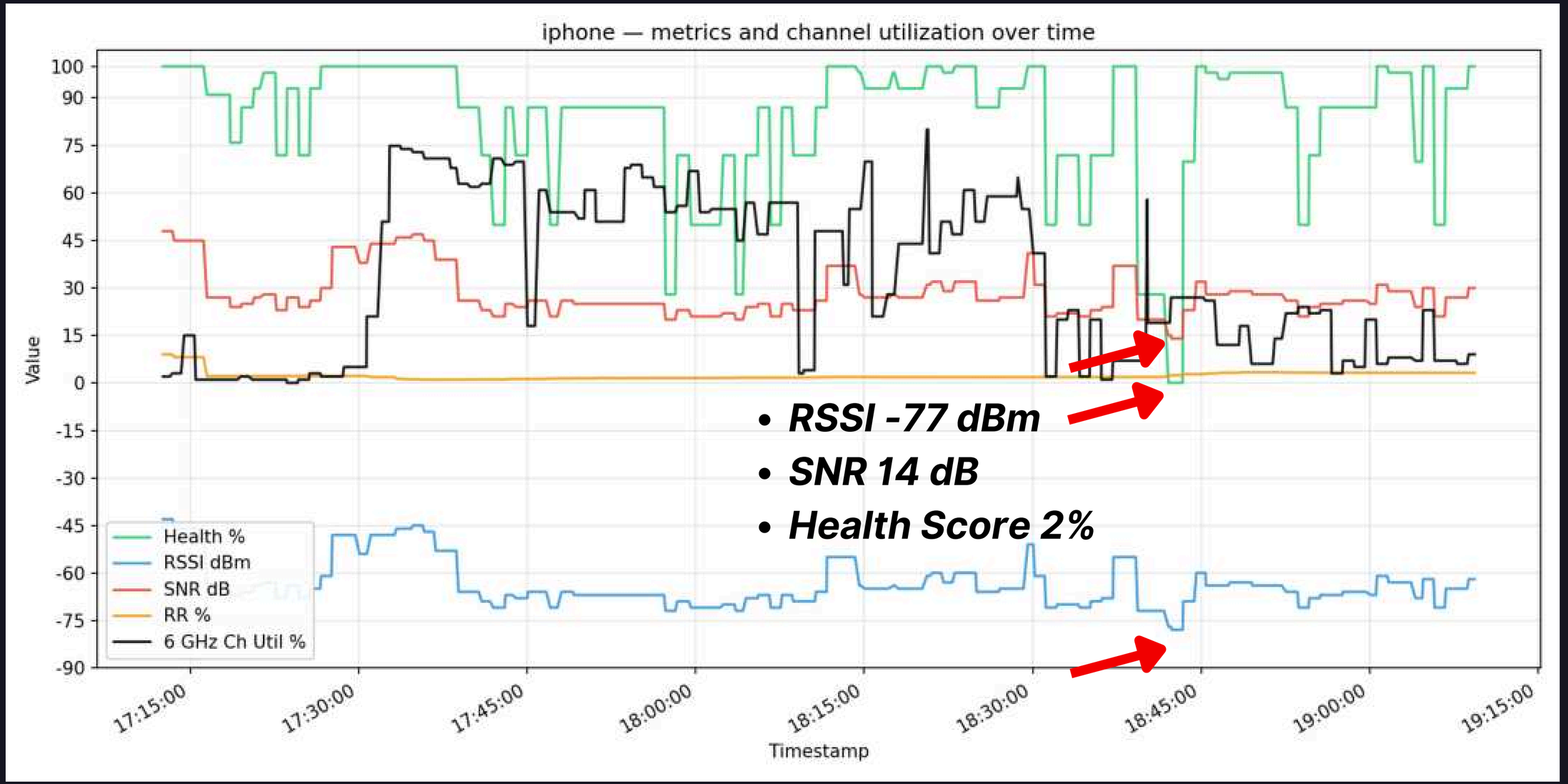
# iPhone 16 Pro Max



What else?

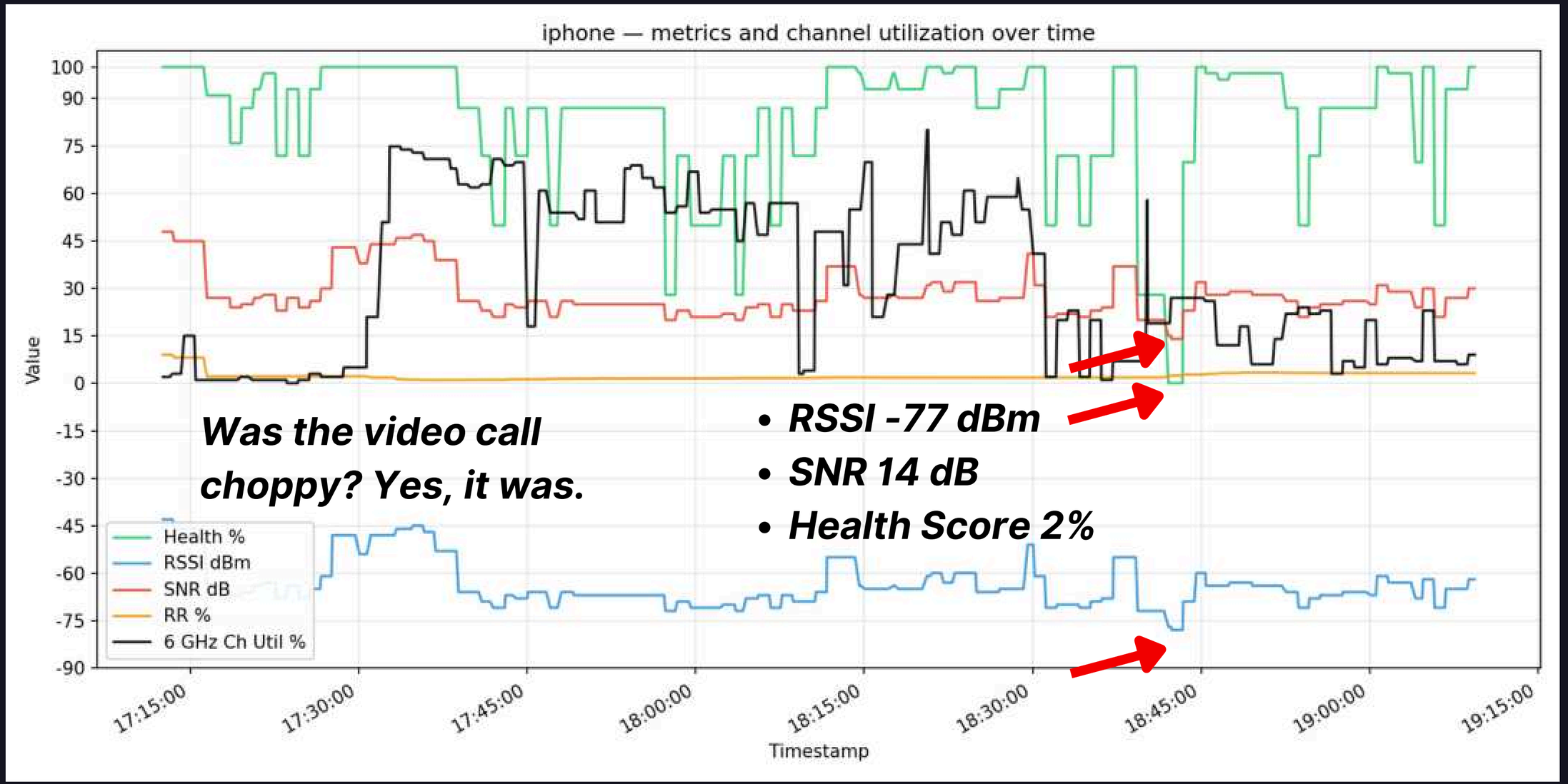
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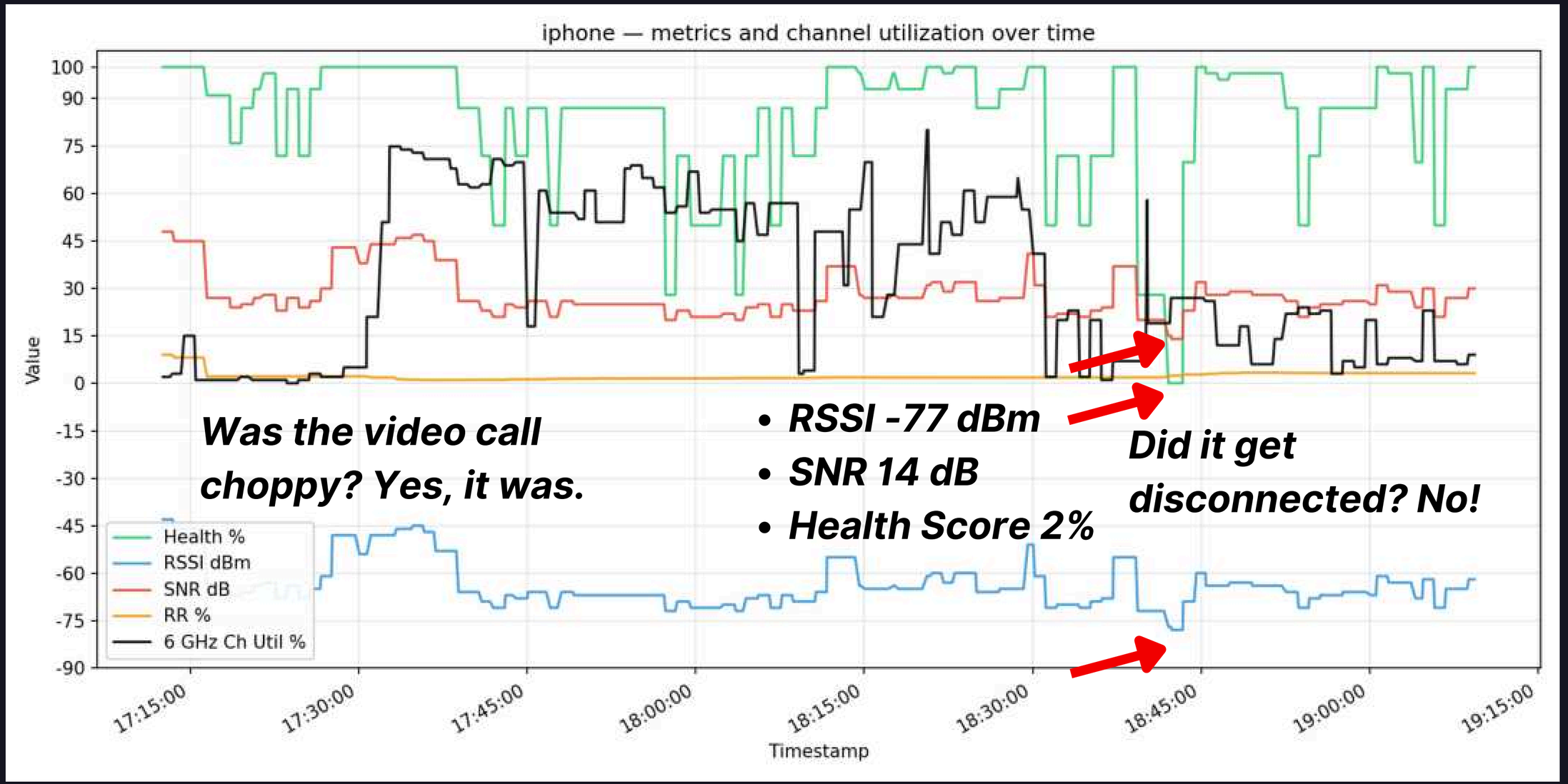
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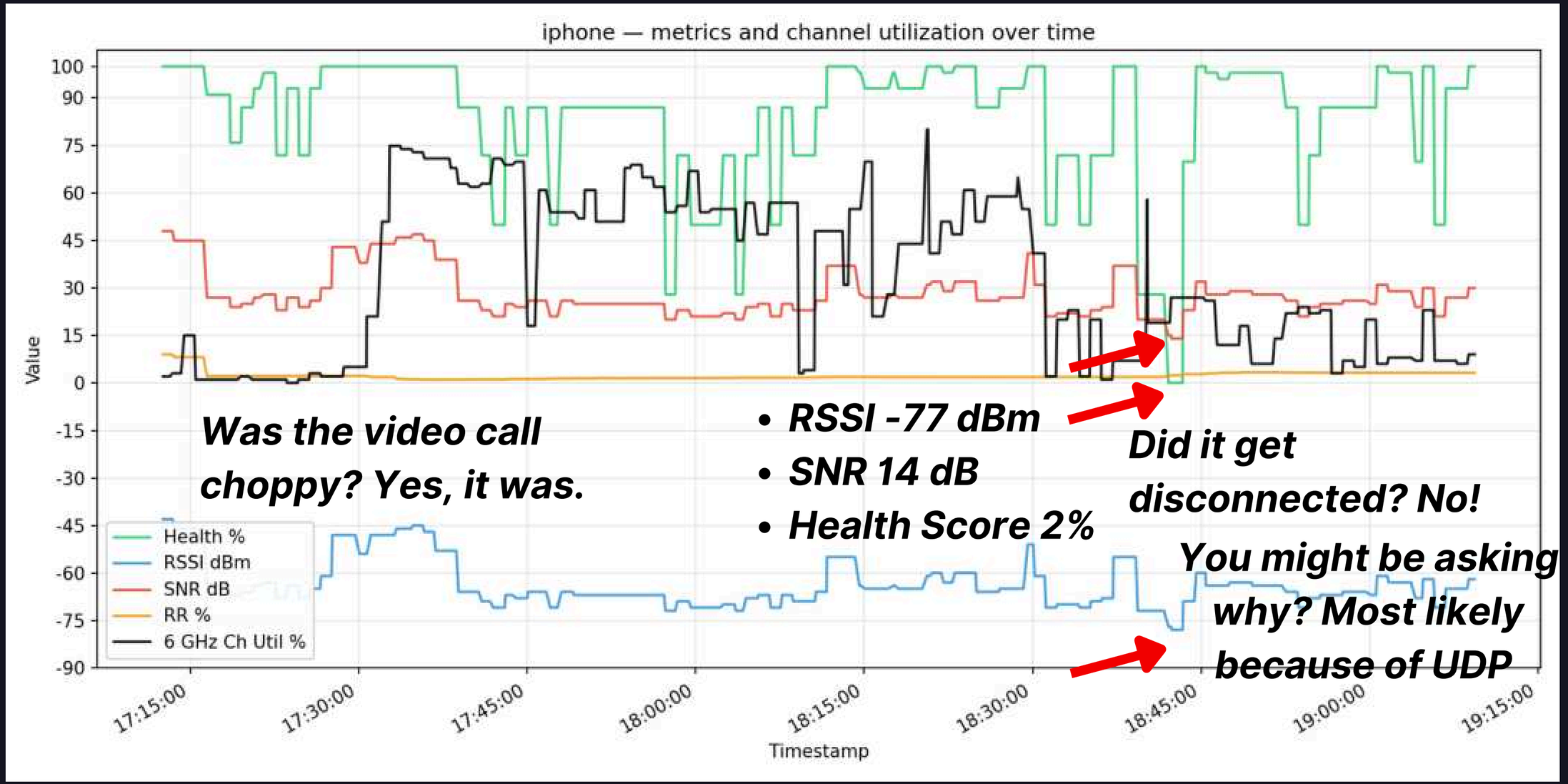
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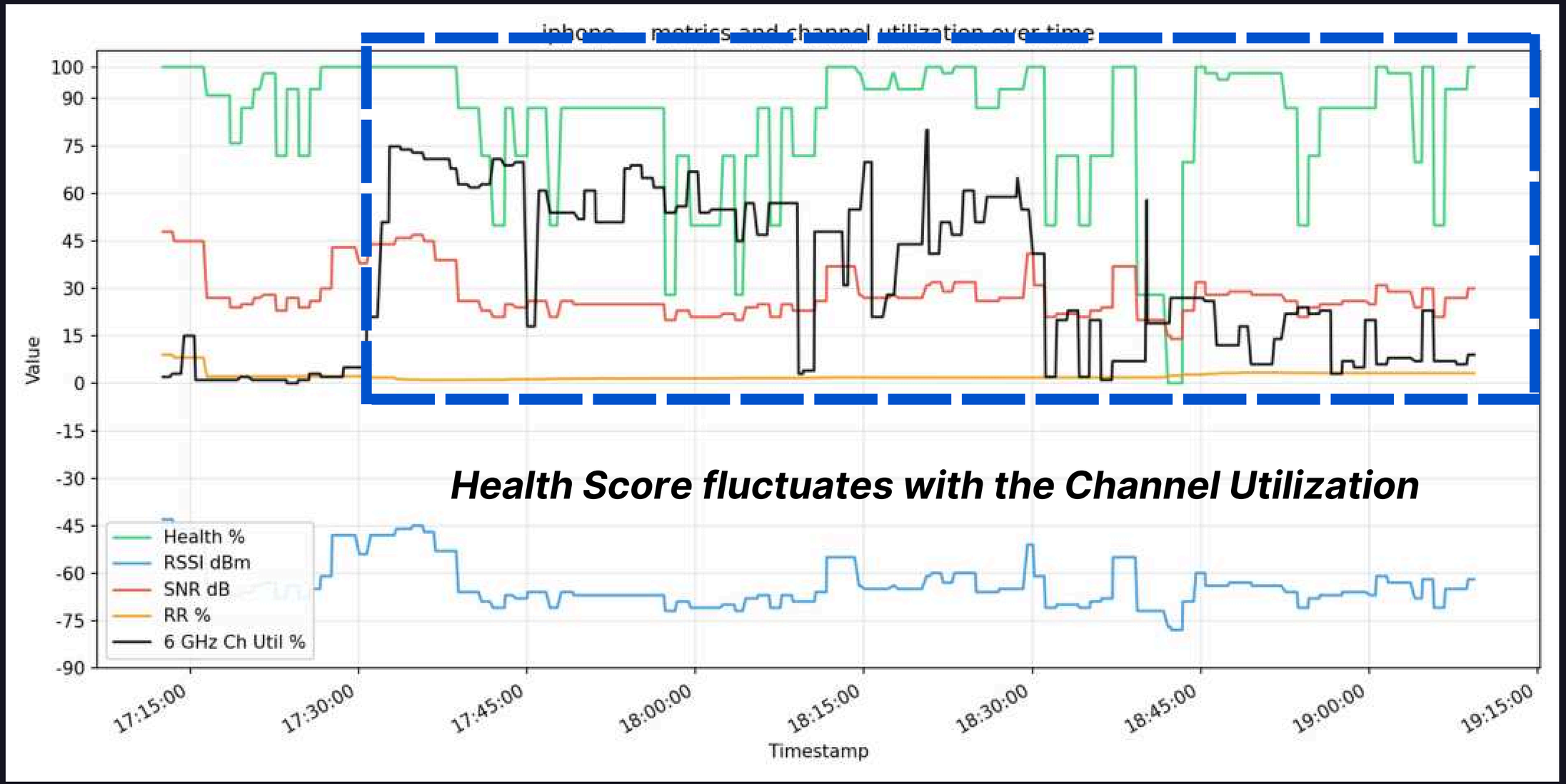
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More?

# iPhone 16 Pro Max





# iPhone 16 Pro Max



# My Learning 4:

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# My Learning 5:

Channel utilization plays an important role in WLAN health, especially for mobile clients. (There’s a value in monitoring Ch Util while validation a WLAN)

# iPhone 16 Pro Max



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There is a strong correlation between RSSI and SNR.

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Channel utilization plays an important role in WLAN health, especially for mobile clients. (There’s a value in monitoring Ch Util while validation a WLAN)

# My Learning 6:

A client may support the latest IEEE 802.11 standard, but the data rate is not always the maximum supported by that standard; it adapts dynamically based on RSSI and SNR. 4K-QAM; I haven't seen it being used by the device that I am using.

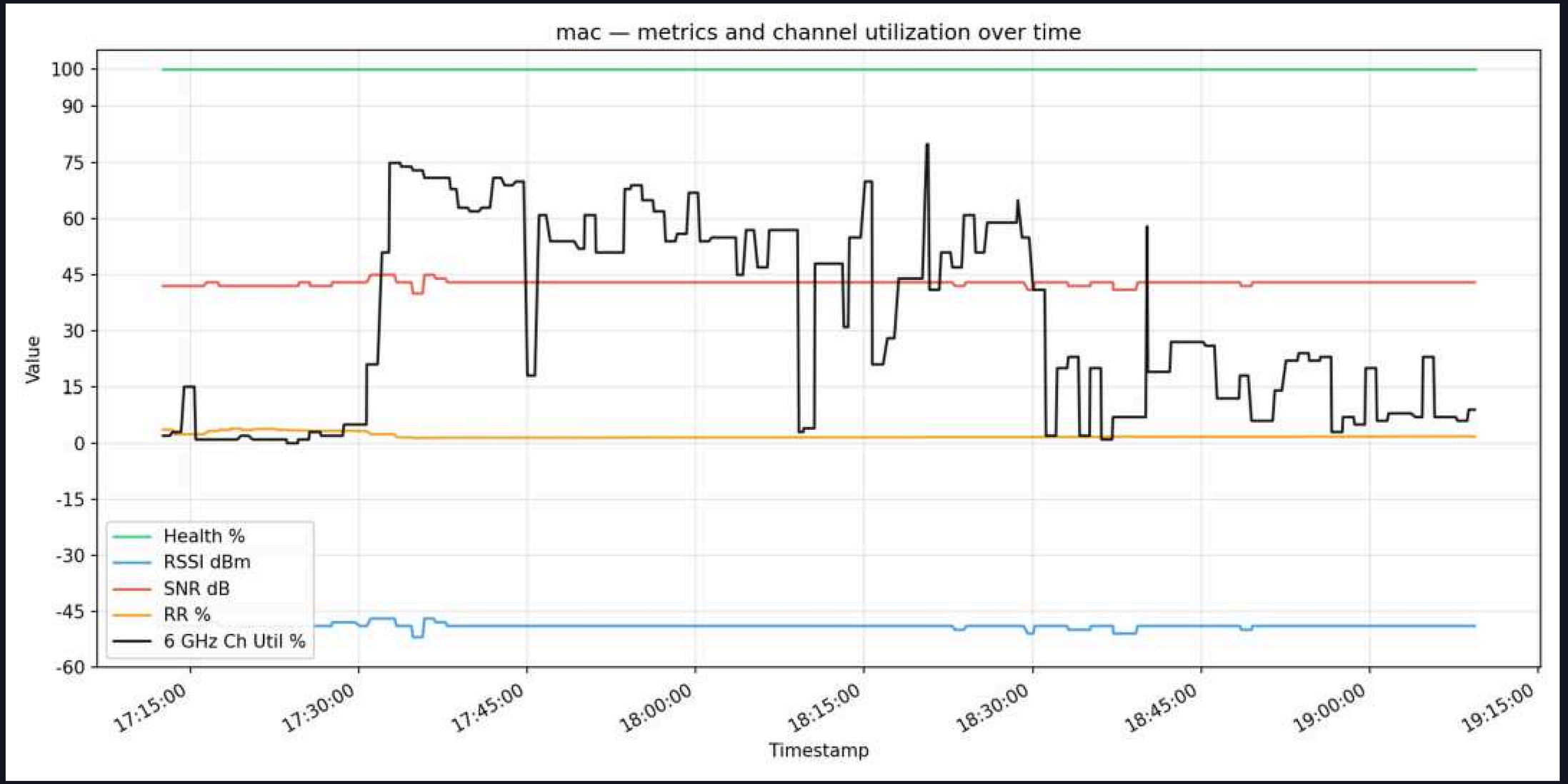
# MacBook Pro 2024



- Was placed next to the AP.
- Spotify and YouTube were playing as well as running speed tests intermittently to generate more traffic.

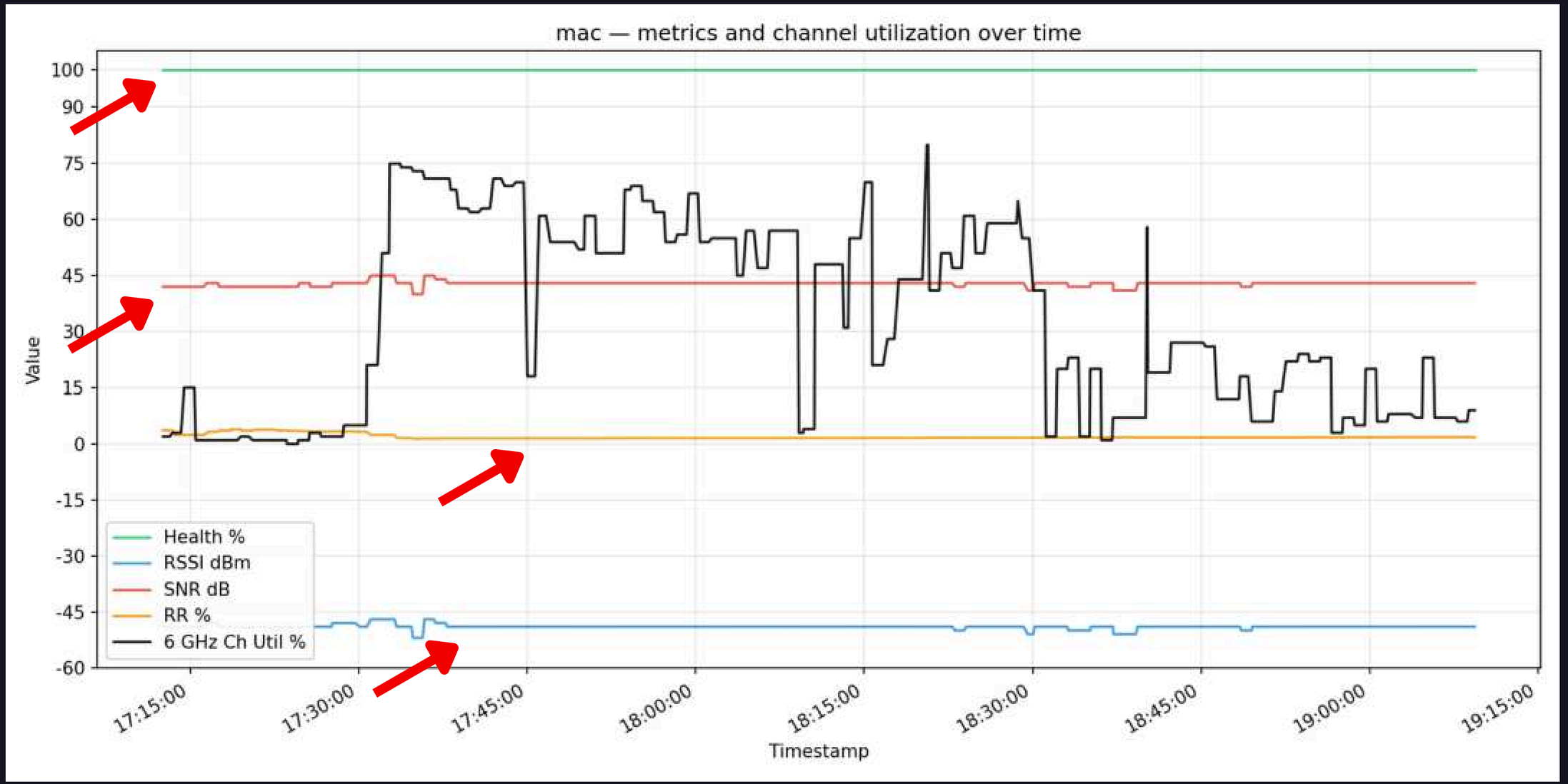
# MacBook Pro 2024





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# My Learning 7:

Distance between a wireless client and the AP is crucial for maintaining service continuity; in general, closer is better!

# MacBook Pro 2024



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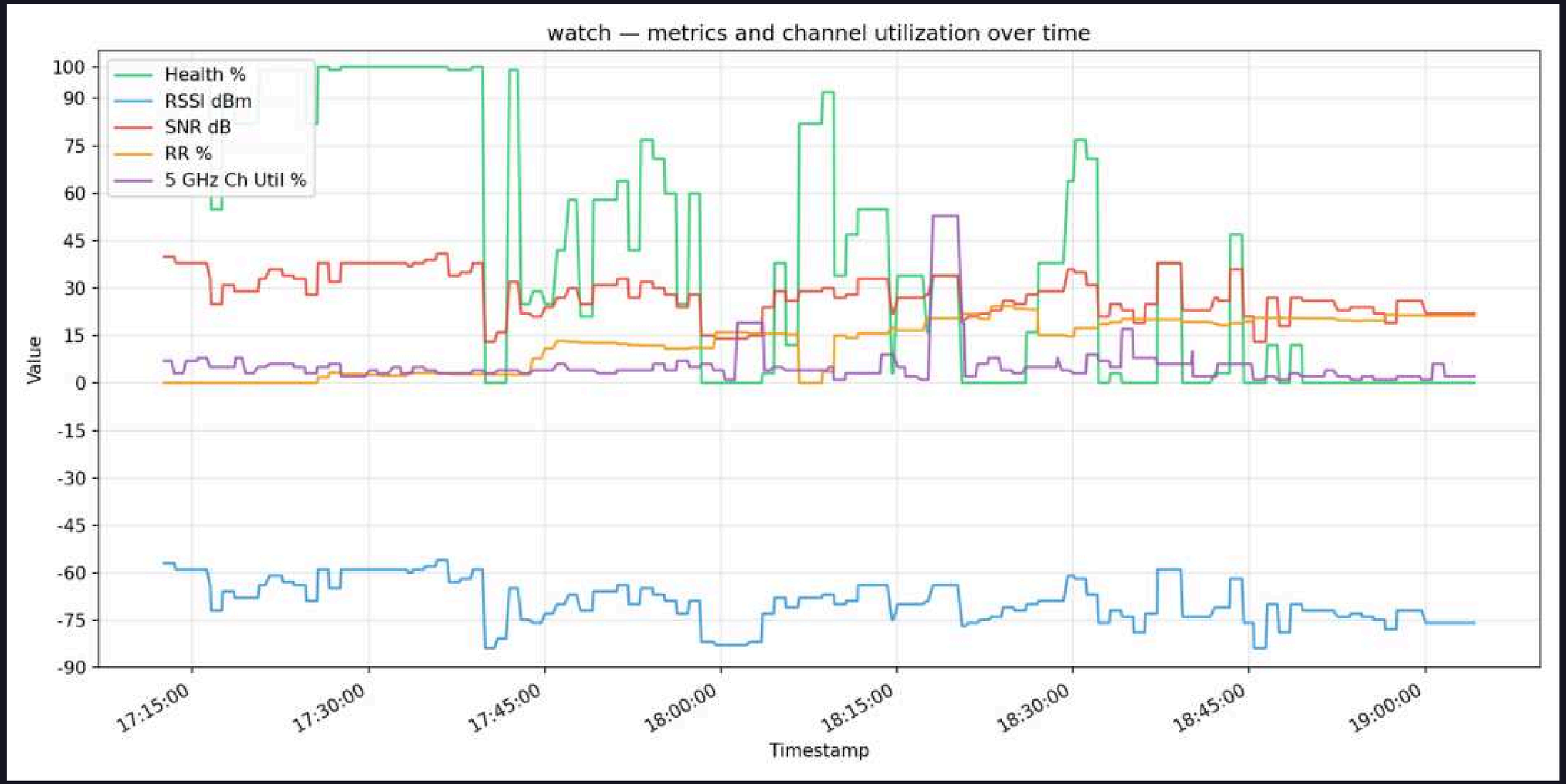
A static client typically shows fewer health fluctuations than a mobile client.

# Apple Watch Series 9



- Was worn by me.

# Apple Watch Series 9



# Apple Watch Series 9 

# My Learning 9:

A client must be actively using the network to accurately evaluate its health condition.

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# Apple Watch Series 9 

# My Learning 9:

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# My Learning 10:

A client may support an IEEE 802.11 standard, but it doesn't necessarily guarantee it supports all of the features. (802.11n client and operating in 20 MHz Channels)

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  **-Apple Watch**
- A client’s capabilities don’t necessarily mean these features will be used.
  **-iPhone 16 Pro Max & Apple Watch**

# Thanks for Listening



Amin Sedighfar
CWNE #622

WLPC Prague
OCT 2026